

Final Year Project

MSc ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Agentic Vision-Language System for Domain-Specific Object Detection and Activity Monitoring in Hospital Ward Environments

By

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Abstract

Hospital wards depend on continuous human observation that is increasingly difficult to sustain at safe quality during night shifts, weekends, and periods of acute understaffing (Haque et al., 2020; Esteva et al., 2021). This dissertation presents the design, implementation, and evaluation of an artificial intelligence based patient activity monitoring system intended to support, not replace, that observation. The system converts a single overhead red-green-blue (RGB) camera frame into a structured nurse-readable description of the patient and the immediate scene, with explicit alerts when safety-critical events are detected. Five layers are stacked into a single pipeline. Layer one is a You Only Look Once version 8 (YOLOv8) detector (Terven et al., 2023) fine-tuned on the hospital_bed Roboflow dataset of 2,004 images and 11 classes that span patient postures, hazards, and ward objects. Layer two is the Common Objects in Context (COCO) pretrained YOLOv8-pose head, providing 17 keypoints per detected person. Layers three and four perform deterministic geometric reasoning over bed-relative position and head orientation. Layer five is a temporal state machine that aggregates per-frame facts over a sliding window to recognise composite activities such as sleeping, drinking, restlessness, and mobilising with a walking aid. Ten experiment configurations spanning three model sizes, three confidence thresholds, and the application or omission of test-time augmentation were compared on validation accuracy of patient activity classification. The winning configuration, YOLOv8 small with confidence threshold 0.25 and no test-time augmentation, reached an overall activity-classification accuracy of 91.04 percent on the 402-image validation split, including 100 percent accuracy on lying, 92 percent on fall, and 95.52 percent on the binary danger-flag classification. Detection mean Average Precision (mAP) at the 0.50 Intersection over Union (IoU) threshold reached 0.9433, and mAP averaged from 0.50 to 0.95 reached 0.7801. The contribution lies in the explicit, layered integration of a fine-tuned hospital-domain detector with a generic pose head and a deterministic temporal aggregator, which prioritises interpretability, traceability, and bedside relevance over end-to-end black-box reasoning.

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1 Introduction

1.1 Context of Research

Hospital wards are dense, fluid, and unforgiving environments. A single nursing team typically watches over multiple patients of varying acuity, and the cost of a missed event, whether that is a fall, a deteriorating posture, or a patient pulling at an intravenous line, can be both clinical and human. Manual observation, which remains the prevailing model in most general wards, depends on staff who are simultaneously charting, dispensing medication, escorting visitors, and answering call bells. The result is uneven coverage that worsens during night shifts and periods of acute understaffing. Litjens et al. (2017) describe how computer vision techniques have begun to reshape clinical workflows by extracting structured information from images that would otherwise demand dedicated human attention, and Esteva et al. (2021) extend this argument to medical computer vision more broadly, highlighting the maturity of deep learning for tasks ranging from anatomical detection to diagnostic triage and procedural support.

Patient monitoring sits between two adjacent fields. The first is human activity recognition (HAR), a well-mapped territory in computer vision research that asks how a system can label what a person is doing from a temporal stream of imagery or sensor data (Vrigkas et al., 2015; Kong and Fu, 2022; Pareek and Thakkar, 2021). The second is medical computer vision, which has historically focused on diagnostic imaging in radiology and pathology rather than on continuous environmental observation. The intersection of the two, ambient intelligence in care settings, has received increasing attention over the past five years. Haque et al. (2020) define ambient intelligence as a system whose value lies in continuously interpreting an environment in support of clinical decisions, reducing burden on staff and surfacing events that human attention is likely to miss. Diraco et al. (2023) survey this domain across smart-living applications and emphasise that the difficulty is rarely raw recognition accuracy on a public benchmark, but rather the integration of recognition into a workflow that clinicians can trust. Ahmed et al. (2022) argue further that the practical viability of such systems rests on edge-friendly deployment so that personal data never leaves the local device.

This dissertation sits squarely in that intersection. It targets a narrowly scoped clinical question. Given a camera positioned to observe a hospital bed, can a system reliably tell a nurse what the patient is doing in the language they would themselves use? The system does not aim to replace observation. It aims to summarise it. The output is a short, structured description per frame containing patient state, fall risk, bed status, side of bed, face direction, mobility support, and any active alerts. Although the underlying machinery is deep, the surface presented to the user is deliberately shallow, because a busy nurse cannot read more than a few lines from a glance at a monitor positioned in a side room.

1.2 Project Aim

The aim of the project is to design and implement a multi-layer perception system that converts overhead RGB frames of a hospital bed into a structured, nurse-readable activity description with explicit safety alerts. The aim is bounded by four load-bearing constraints, each of which shapes every design decision that follows:

- Real-time inference. The end-to-end pipeline must process frames at no fewer than ten frames per second on a consumer-grade graphics processing unit (GPU), which rules out architectures that require a forward pass through a heavyweight transformer on every frame.
- Open-source dependencies. Only open-source models, datasets, and libraries may be used, so that the artefact is reproducible and redistributable under permissive licenses.
- Traceability. Every safety-critical alert must be reducible to a specific frame-level fact that a clinician can audit, eliminating the kinds of opaque reasoning that would not survive clinical governance review.
- Narrow scope over broad scope. Ward-level patient observation is preferred over open-domain hospital scene understanding, because the clinical value of a narrow, reliable system exceeds that of a broad, brittle one in a setting where false alarms erode trust faster than missed alerts.

1.3 Project Objectives

The project is decomposed into six objectives, with novelty residing in their integration rather than in any single component:

- **Objective 1: Hospital-specific detection:** Fine-tune YOLOv8 (nano/small/medium) on the hospital_bed Roboflow dataset to recognise patient postures, ward bed, mobility aids, and explicit hazards from a single frame, balancing accuracy with inference latency (Terven et al., 2023; Jiang et al., 2022).
- **Objective 2: Generic pose estimation:** Add a skeletal-geometry layer so posture and head orientation can be inferred from joints rather than pixels, transferring the operating-theatre approach of Chen et al. (2018) to the bedside.
- **Objective 3: Deterministic geometric reasoning:** Hand-coded rules operating on bounding boxes and keypoints to infer the patient's spatial relationship with the bed and face orientation — relations not derivable from class-level detection alone.
- **Objective 4: Temporal state machine:** Aggregate frame-level facts over a sliding window to express composite activities (sleeping, drinking, restlessness), since single-frame reasoning is inherently brittle (Vrigkas et al., 2015; Pareek and Thakkar, 2021).
- **Objective 5: End-to-end orchestrator:** Combine all layers into a single per-frame entry point returning a flat dictionary for logging, dashboards, and alerting, implemented as the MonitorPipeline class in pipeline.py.
- **Objective 6: Quantitative and qualitative evaluation:** Evaluate on the supplied dataset and held-out frames with explicit error analysis, reported in chapters 7 and 8.

Unlike category-only detection work (Zhao et al., 2019; Zou et al., 2023), keypoint-only pose pipelines (Chen et al., 2018), wearable-dependent HAR (Mukherjee et al., 2020), infrastructure-heavy multi-view systems (Singh et al., 2019), or opaque VLM approaches (Wang et al., 2024; Liu et al., 2024), this system pairs a small fine-tuned detector with a general pose head and deterministic temporal aggregator to produce

nurse-readable JSON in which every alert is traceable to a frame-level fact e.g., a *sleeping* label requires ≥85% of recent frames to show lying-on-bed, low motion, and a high fraction of unregistered eye keypoints, with no silent overrides.

2 Literature Review

2.1 Research Background and Definitions

Human activity recognition is the task of producing a label that describes what a person is doing from one or more sensory inputs, typically video or wearable sensors. Vrigkas et al. (2015) propose a foundational taxonomy that distinguishes simple actions, interactions, group activities, and behaviours, and they observe that the difficulty of the task scales with the temporal scope of the label. A wave of a hand can be classified from a fraction of a second of video; recognising that a patient is restless across a night requires aggregating evidence over a much longer interval. Kong and Fu (2022) update this taxonomy in light of deep learning, noting that the dominant modern paradigms are end-to-end convolutional or transformer architectures applied to spatiotemporal volumes. Pareek and Thakkar (2021) survey video-based HAR specifically, cataloguing benchmark datasets and discussing the open challenges of cluttered backgrounds, occlusion, and viewpoint variance. Gupta et al. (2022) take a complementary narrative-review angle, arguing that the field has matured beyond benchmark accuracy into questions of generalisation, data efficiency, and ethical deployment.

Object detection is a closely related but distinct task. Where HAR asks what is happening, object detection asks what is present and where. Zhao et al. (2019) trace the deep-learning era of object detection from Region-based Convolutional Neural Network (R-CNN) through Single Shot Detector (SSD) and YOLO, organising the literature around the two-stage versus single-stage axis. Zou et al. (2023) cover an even longer arc of object detection across two decades, showing how single-stage detectors have come to dominate real-time deployment. The YOLO family in particular (Redmon and Farhadi, 2018; Bochkovskiy et al., 2020; Terven et al., 2023; Jiang et al., 2022) has become the workhorse of practical detection systems because of its consistent accuracy-speed trade-off and its mature open-source ecosystem.

Pose estimation, the third primitive used in this work, operates between detection and recognition by predicting the locations of anatomical landmarks on each detected person. The COCO 17-keypoint convention adopted by YOLOv8-pose covers facial landmarks,

shoulders, elbows, wrists, hips, knees, and ankles, providing sufficient geometry for posture inference and for proxies of behaviour such as hand-to-face proximity. Chen et al. (2018) demonstrate that pose estimation can be specialised to clinical environments without losing the generality that makes the COCO models useful, showing strong qualitative performance on a patient-pose dataset captured in operating theatres. The result that pose representations transfer to clinical scenes more robustly than appearance representations is one of the reasons this project keeps the pose head COCO-pretrained rather than fine-tunes it.

2.2 Similar Works

Deep learning provides the backbone for every layer in this project, and it is worth reviewing both its general architecture and its application to medical imagery. Alzubaidi et al. (2021) provide a comprehensive review of convolutional neural network (CNN) architectures, organising them by depth, residual connections, attention mechanisms, and efficiency considerations. Their treatment of MobileNet-style and EfficientNet-style backbones is directly relevant to the question of which YOLOv8 size to deploy at the bedside, where energy and latency constraints rule out the heaviest backbones. Litjens et al. (2017) survey deep learning specifically in medical image analysis, noting that the field has settled into a small number of canonical task formulations: classification, detection, segmentation, registration, and generation. Minaee et al. (2022) review image segmentation in particular, which is informative for any future extension of this work toward instance-level masks of bed and patient that would refine the bed-relative spatial reasoning currently performed on bounding boxes alone.

The YOLO lineage is central to the detection layer of this system. Redmon and Farhadi (2018) introduced YOLOv3 with the multi-scale prediction head that has informed every subsequent version. Bochkovskiy et al. (2020) extended the lineage with YOLOv4 and the closely related YOLOv7, demonstrating that careful engineering of the anchor head, the neck, and the data augmentation strategy could meaningfully improve mAP without architectural revolution. Jiang et al. (2022) review the developments through YOLOv5 and emphasise the role of training-time augmentations, while Terven et al. (2023) provide the most comprehensive comparison through to YOLOv8 and YOLO Neural Architecture

Search (NAS), including detailed inference-time benchmarks that informed the model-size choice in this project. The pattern that emerges from these reviews is consistent. YOLOv8 represents a thoughtful consolidation rather than a revolution, with the anchor-free head, decoupled detection, and improved augmentation defaults that make fine-tuning on small domain datasets reliable.

Object detection in clinical scenes is less commonly surveyed than open-domain detection. Zhao et al. (2019) and Zou et al. (2023) both note that medical applications skew the metrics that matter. False negatives on safety-critical classes are more costly than false positives, and confusion among visually similar classes (such as lying versus sleeping) often matters less than the confusion between safety-critical and non-safety-critical classes (such as fall versus lying). The same logic motivates the activity-priority rule used in chapter 6 of this work, where fall is given precedence over all other activity labels.

Pose estimation in clinical environments has a smaller but instructive literature. Chen et al. (2018) demonstrate that COCO-pretrained pose networks transfer to operating-theatre environments with relatively modest fine-tuning, supporting the choice in this project to use the stock YOLOv8-pose model rather than fine-tune one. Their work also motivates the architectural decision to keep the pose layer separate from the detection layer rather than fold them into a single multitask network. The failure modes are distinct, and a separate failure mode is a debuggable failure mode.

Activity recognition outside the clinical domain has been explored extensively with both wearable and visual sensors. Akbari et al. (2020) propose a hierarchical signal segmentation strategy that segments a continuous Inertial Measurement Unit (IMU) stream into activity bursts before classifying each burst. The hierarchical structure is conceptually similar to the temporal state machine used in this project, where frame-level facts are aggregated before composite activity labels are emitted. Mukherjee et al. (2020) show that an ensemble of compact CNNs trained on smartphone IMU data can reach high accuracy on standard HAR benchmarks, but they also highlight the fundamental constraint that wearable HAR requires explicit instrumentation of every patient. The bedside camera scenario in this project has the opposite trade-off. No instrumentation is

required, but the visual signal is more ambiguous and demands greater downstream reasoning. Singh et al. (2019) approach activity recognition from multi-view fused features, achieving strong results on standard surveillance datasets. Their work is closer in spirit to ours, but it requires multiple synchronised cameras, which is uncommon in routine ward installations.

Ambient intelligence in healthcare has been argued for explicitly by Haque et al. (2020), who frame the deployment of continuous environmental sensing in care settings as a public-health imperative rather than a research curiosity. Esteva et al. (2021) survey the applications of deep learning in medical computer vision more broadly and identify ward-level monitoring as one of the under-served frontiers compared with diagnostic imaging. Diraco et al. (2023) catalogue smart-living HAR systems and emphasise the issues of multimodality, interoperability with clinical information systems, and resource-constrained processing. Their resource-constrained processing argument supports the design decision in this project to favour a small fine-tuned detector over a heavyweight VLM at the perception layer.

Vision-language models occupy an adjacent corner of the design space. Radford et al. (2021) introduced Contrastive Language-Image Pre-training (CLIP), a contrastive image-text dual encoder that enabled zero-shot classification by aligning visual and textual representations. Li et al. (2023) extended this paradigm with Bootstrapping Language-Image Pre-training 2 (BLIP-2), which bridges frozen image encoders and large language models through a lightweight Q-Former, dramatically reducing the cost of training open-domain visual question answering systems. Liu et al. (2024) developed visual instruction tuning, which produced the Large Language and Vision Assistant (LLaVA) family of models, and Wang et al. (2024) released Qwen2-VL with high-resolution dynamic processing that has set new public benchmarks on visual question answering and document understanding. These systems offer end-to-end scene reasoning with rich linguistic output, which is appealing for clinical applications but raises serious concerns about latency, hallucination, and traceability. The decision in this project to use deterministic geometric and temporal reasoning rather than a VLM was made on those grounds. A rule that fires when bed Intersection over Union is at least 0.15 can be

inspected, debugged, and audited. A VLM hallucination of “the patient is comfortable” cannot.

Person identification, while not implemented in the deployed pipeline, is part of the expected future-work envelope. Schroff et al. (2018) introduced FaceNet, the triplet-loss-trained face embedding system that defined the modern face-recognition pipeline. Deng et al. (2022) extended this with ArcFace, whose additive angular margin loss has become the de facto standard for deep face verification. Both would be candidates for the optional re-identification layer originally specified in the project brief, and both raise consent concerns that are addressed in chapter 3 of this dissertation.

Augmentation is a separate but crucial concern in any small-dataset domain transfer. Shorten and Khoshgoftaar (2019) review the breadth of image-augmentation techniques used in deep learning and argue that augmentation is rarely a single design choice but a collection of overlapping transforms whose value depends on the failure modes of the downstream task. The augmentation choices made in chapter 6, including hue-saturation-value (HSV) jitter, random translation, and mosaic mixing, are directly informed by their analysis.

Ambient assistive robotics provides the broader context for ambient monitoring. Martinez-Martin and del Pobil (2018) review robot assistants for elderly care, identifying continuous activity awareness as the central enabler of any assistive function, whether the actuator is a robot, an alerting system, or a logging dashboard. Their argument supports the position taken in this dissertation that activity awareness should be considered a primitive on which more elaborate care-support systems are built rather than an end in itself. Wu et al. (2021) survey graph neural networks, which are increasingly used to model relations among detected entities. A graph-based extension of the spatial reasoning layer is one of the candidate future directions discussed in chapter 9.

A point worth raising explicitly concerns the metrics under which the literature compares object detectors. Most public benchmarks report mAP at the 0.50 IoU threshold (mAP at 0.50) and the average over the 0.50 to 0.95 IoU range (mAP at 0.50 to 0.95), the latter being the more demanding metric because it penalises loose localisation. Zhao et al.

(2019) and Zou et al. (2023) note that medical detection tasks often care less about strict localisation than about classification correctness, particularly when the bounding box is downstream of pixel-accurate segmentation. The clinical decision is rarely “where exactly is the bed” but “is the patient on the bed”, which is governed by an IoU threshold that tolerates moderate localisation error. The choice in this project to set the on-bed threshold at IoU at least 0.15, well below conventional benchmark thresholds, reflects this distinction. The same observation explains why the headline metric in chapter 7 is activity-classification accuracy rather than mAP. Detection metrics measure intermediate quality. The clinical user reads activity labels, not boxes.

Edge-deployment perspectives on smart healthcare add a final piece to the literature backdrop. Ahmed et al. (2022) demonstrate that a deep-learning patient-discomfort detection system can be made to run at the network edge by careful model selection and quantisation, and they argue that this architecture is the only credible path to clinical adoption that respects data minimisation. The implication for this dissertation is that the design must remain compatible with edge deployment from the outset. A bedside camera that streams raw video to a cloud inference server is unlikely to receive consent in any modern hospital governance framework, while a camera with on-device inference can be defended through local processing alone. The choice of small backbones in the experiment suite of chapter 7 is partly motivated by this constraint. A YOLOv8 small with confidence threshold 0.25 is small enough to run on a Jetson-class device and large enough to retain the activity-classification accuracy required for clinical relevance.

2.3 Summary

The literature across HAR, object detection, pose estimation, ambient intelligence, and vision-language modelling converges on a single observation that frames the design choices in this dissertation. Open-domain accuracy on benchmarks is no longer the main constraint on practical deployment. The constraints that matter are interpretability, data efficiency for narrow domains, traceability of safety-critical decisions, and runtime feasibility on unattended hardware. The five-layer pipeline described in chapters 5 and 6 is shaped by those constraints. A small fine-tuned YOLOv8 satisfies the data-efficiency and runtime requirements; a generic pose head satisfies geometric expressiveness

without requiring more data; a deterministic spatial reasoning layer satisfies traceability; a temporal state machine satisfies the need for composite activity labels at clinically relevant time scales. None of these primitives is novel in isolation; the contribution is in their composition for the specific case of bedside patient monitoring, with explicit nurse-readable output that grounds every alert in a sequence of frame-level facts.

3 Methodology

3.1 Data Collection

The dataset used to train the activity detector is the hospital_bed Roboflow dataset, a community-contributed collection of 2,004 annotated hospital images covering 11 classes. The class list is read directly from the dataset configuration file at the start of the notebook:

```
ACTIVITY_CFG['names']
# ['bed', 'danger', 'fall', 'lie', 'move', 'patient',
#  'person', 'sit', 'stand', 'walker', 'wheelchair']
```

The dataset is split into 1,401 training images, 402 validation images, and 201 test images. The export format is YOLOv8 native, with one text label file per image, where each line records a class index followed by normalised bounding-box coordinates. The original Roboflow export uses relative paths in the data.yaml file, which were rewritten with absolute paths during environment setup so that the Ultralytics framework resolves splits correctly when the training process is invoked from a working directory other than the dataset folder.

The choice of this dataset over the larger Hospital Indoor Object Detection (HIOD) dataset originally proposed in the project specification was made for two reasons. First, HIOD is gated behind an authorial request that did not return within the project timeline. Second, the hospital_bed dataset is more tightly scoped to bedside-level patient monitoring rather than general hospital scene understanding, which aligns more cleanly with the scope of this project. The supplementary hospital equipment dataset was retained for exploratory data analysis but was not used to train any deployed model, because its 25 classes (anaesthesia machines, C-arms, and similar operating-theatre equipment) are not relevant at the bedside.

3.2 Data Analysis

Exploratory data analysis confirmed three findings that shaped the training strategy. The class distribution is heavily skewed: bed, patient, and person dominate the training

instances while fall, danger, walker, and wheelchair appear in tens or low hundreds of instances. This imbalance is realistic, since real wards contain many beds and few falls, but it constrains how much can be expected from minority classes without targeted oversampling. Images per split are roughly stable, with no split missing any class except the test split, which contains only fall and danger images and was therefore unsuitable as the primary evaluation set for activity classification. The example images show typical ward scenes captured at moderate viewing angles, with one bed and one patient in the dominant frame.

These findings led to three decisions. First, the validation split was promoted to be the primary evaluation set because it preserves all six activity classes (the five activity labels plus a no-detection bin). Second, no aggressive minority-class oversampling was applied, because the YOLOv8 default loss weighting handles moderate imbalance reasonably and oversampling can amplify label noise on rare classes (Shorten and Khoshgoftaar, 2019). Third, default Ultralytics augmentations were retained, because the prevailing literature on small-dataset domain transfer (Shorten and Khoshgoftaar, 2019; Litjens et al., 2017) finds them robust, and the project timeline did not allow for an augmentation ablation.

3.3 Model Selection

YOLOv8 was selected over alternative real-time detectors for three reasons. First, the YOLOv8 family covers a wide spectrum of model sizes from nano (3 million parameters) through small, medium, large, and extra-large, allowing the size to be tuned to the available hardware after the initial fine-tune is complete (Terven et al., 2023). Second, the Ultralytics framework that ships YOLOv8 also provides a compatible YOLOv8-pose model for keypoint estimation, which removes the integration overhead of mixing two distinct frameworks. Third, the family has consistent inference latency characteristics across sizes, which makes the size-versus-accuracy trade-off explicit and tunable at deployment time (Jiang et al., 2022).

The model-size sweep covered nano, small, and medium backbones, all fine-tuned for 30 epochs with seed 42, a batch size of 32 for the nano variant and 24 for small and medium to fit within video random-access memory (VRAM), and image resolution 640 by

640. Optimiser selection was left to the framework's automatic mode, which picked Adaptive Moment Estimation with Decoupled Weight Decay (AdamW) at the default learning rate of 0.01 with cosine-free decay, momentum 0.937, weight decay 0.0005, and three warmup epochs. These hyperparameters reflect the training defaults documented in the Ultralytics framework and are retained without modification, because each modification would have demanded its own ablation, and the experiment budget was prioritised for the size, confidence, and test-time augmentation sweep instead.

3.4 Project Planning

3.4.1 Initial Plan

The initial project plan, agreed with the supervisor at the start of the academic year, proposed a five-phase delivery over fifteen weeks. Phase one (weeks one to three) would conduct the literature review and select an open-source vision-language model from candidates including LLaVA, Qwen2-VL (Wang et al., 2024), and InternVL. Phase two (weeks three to six) would request and obtain the HIOD dataset, prepare supplementary annotations for activity-related objects such as water bottles and food trays, convert annotations to the YOLO format, and fine-tune YOLOv8 with benchmark comparison against published HIOD results. Phase three (weeks five to nine) would design the structured detection-context builder, implement the agentic VLM reasoning pipeline, and add person re-identification using FaceNet (Schroff et al., 2018) or ArcFace (Deng et al., 2022). Phase four (weeks eight to eleven) would integrate the layers, build a Streamlit or Gradio dashboard, and implement alert mechanisms. Phase five (weeks ten to fifteen) would evaluate, iterate, and document.

3.4.2 Actual Plan

Task	Wk1	Wk2	Wk3	Wk4	Wk5	Wk6	Wk7	Wk8	Wk9	Wk10	Wk11	Wk12	Wk13	Wk14	Wk15	Wk 16
Literature Review	■	■	■	■	■	■	■	■	■							
HIOD Dataset Request & Setup	■	■														
Dataset Preparation & Conversion			■	■												
YOLOv8 Fine-tuning on HIOD				■	■	■										
Detection Layer Evaluation						■	■									
VLM Selection & Setup					■	■										
Agentic Reasoning Pipeline Dev							■	■	■							
Person Re-ID Integration									■	■						
System Integration								■	■	■	■					
Dashboard Development										■	■	■				
Testing & Evaluation											■	■	■			
Dissertation Writing							■	■	■	■	■	■	■	■	■	■
Final Review & Submission Prep																

The plan that was actually delivered diverged from the initial plan in three ways. First, the HIOD dataset request was not granted within the project window, and the activity-tuned hospital_bed Roboflow dataset was substituted as the primary training source. This was

not a setback but a refocusing, because hospital_bed is more directly aligned with the bedside-monitoring scope of this project than HIOD's 56-category hospital-wide vocabulary. Second, the agentic VLM reasoning layer was replaced by a hand-engineered geometric reasoning layer plus a temporal state machine. The substitution was made for three reasons. The latency budget of 10 frames per second on a consumer GPU is not consistently achievable with current open-source VLMs (Liu et al., 2024; Wang et al., 2024); VLM outputs are not reliably traceable to specific input pixels, which conflicts with the clinical traceability requirement; and the activity vocabulary in scope is small enough that explicit rules cover it more reliably than free-form text generation. Third, the Streamlit or Gradio dashboard was descoped to a programmatic JSON output and an OpenCV overlay generator inside the notebook, partly for time and partly because the JSON output is the durable contract; a dashboard is a presentation layer on top of it and can be assembled by any subsequent project without further changes to the core pipeline. The face-based re-identification layer was deferred to future work for reasons of consent and privacy that are detailed in section 3.5. Despite these changes, all six original objectives were met or exceeded in the dimensions that matter most for clinical relevance: activity recognition accuracy, latency, traceability, and ethical posture.

3.5 Ethical Considerations

The project uses only publicly available datasets and processes no real patient data at any stage. No clinical environments were visited, no consent was sought from any individual, and no personally identifiable information was retained beyond the local session in which an image was processed. This posture aligns with the General Data Protection Regulation (GDPR) principles of data minimisation and purpose limitation. Person re-identification was deferred to future work specifically because deploying any face-recognition component, even one that operates only on locally computed embeddings, raises a higher consent bar than activity-only monitoring. The original brief requirement to integrate ArcFace or FaceNet (Deng et al., 2022; Schroff et al., 2018) is therefore deliberately left as a candidate for future controlled-environment work where consent and governance can be established before, not after, the technical integration. Outputs from the deployed pipeline are advisory only; clinical responsibility remains with

the nursing team, and any deployment in a real ward would require independent ethical review that is out of scope for this MSc dissertation.

3.6 Professional Issues

The British Computer Society (BCS) Code of Conduct identifies competence, integrity, and public interest as the three pillars of professional practice. Each is addressed explicitly in the design of this system. Competence is addressed through the layered architecture that exposes the failure modes of each component to inspection, allowing a knowledgeable user to attribute any incorrect output to the layer that produced it rather than to a single opaque model. Integrity is addressed through the deliberate scoping of the system as decision support rather than diagnosis, and through the documentation of limitations in chapter 8 of this dissertation. Public interest is addressed through the open-source dependency tree (Ultralytics YOLOv8, PyTorch, OpenCV) and the publication-suitable artefact set in the model directory, which would allow any subsequent project to reproduce the trained model from the saved weights and the supplied configuration. The system carries no explicit warranty and would require extensive independent validation before deployment in a regulated clinical environment.

4 Requirements

4.1 Requirement Gathering

Requirements were gathered through three sources. The first was the project specification supplied at the start of the academic year. The second was the set of supervisory meetings held fortnightly via Microsoft Teams during the implementation period from January to May 2026. The third was a structured review of the relevant academic literature that informed both the choice of layered architecture and the choice of evaluation metrics. The clinical relevance of each requirement was verified against the patient-monitoring framing in Haque et al. (2020), Esteva et al. (2021), and Diraco et al. (2023). Constraints on real-time performance were derived from Ahmed et al. (2022), whose smart-healthcare edge-deployment perspective sets a useful lower bound on the achievable latency for ambient sensing systems. Where supervisor and literature priorities conflicted, the literature was treated as advisory and the supervisor's decision was treated as binding, on the principle that an MSc project should reflect the judgement of its examiner.

4.2 Organization of Requirements

Requirements are organised under the Must, Should, Could, Won't (MoSCoW) prioritisation scheme, where Must-have requirements define the minimum viable product, Should-have requirements add clinical value but do not block delivery, Could-have requirements would broaden the system's usefulness in subsequent iterations, and Won't-have requirements are explicitly out of scope for this MSc project. Functional requirements describe what the system does; non-functional requirements describe how it does it. The two are presented separately in the tables below. No Won't-have requirements are listed, because the brief in Manoj.doc already excludes a large set of out-of-scope features such as electronic health record integration and multi-camera fusion, and listing them again here would offer no further information to the examiner.

4.3 Functional Requirements

Table 4.1 below lists the ten functional requirements that the system must satisfy.

ID	Description	Priority	Justification
FR1	Detect patient activity classes (lie, sit, stand, move, fall) from a single RGB frame	Must	Core project goal; without it the system cannot summarise patient state.
FR2	Identify the bed-relative position of the patient (on bed, off bed, side of bed)	Must	Bed status is the most clinically informative spatial fact in ward monitoring (Haque et al., 2020).
FR3	Recognise hazard and danger indicators in the scene as a binary flag	Must	Safety-critical events must surface to the nurse independently of the activity label.
FR4	Aggregate per-frame facts over a sliding window to produce composite activity labels (sleeping, drinking, restless-in-bed, mobilising-with-aid)	Must	Single-frame reasoning cannot distinguish lying-and-sleeping from lying-and-reading (Pareek and Thakkar, 2021).
FR5	Estimate skeletal keypoints for each detected person at a confidence threshold suitable for downstream geometric reasoning	Must	Pose is the input to posture inference, head-direction inference, and the drinking proxy (Chen et al., 2018).
FR6	Estimate head pose to a coarse five-class label (frontal, turned-left, turned-right, face-down, unknown)	Should	Side of bed is most clinically actionable when paired with face direction for pressure-care tracking.
FR7	Detect cup, bottle, or wine glass as a precondition for the drinking activity	Should	Drinking detection is brittle without an object-level cue, and cups are well represented in COCO.
FR8	Detect mobility aids (walker, wheelchair) as preconditions for the mobilising-with-aid label	Should	Distinguishes safe ambulation from unsafe ambulation.
FR9	Emit a structured nurse-readable summary per frame with patient state, fall risk, bed status, side of bed, face direction, mobility support, repositioning need, nurse action, and any active alerts	Must	This is the deliverable surface; without it no clinical user can consume the system's reasoning.
FR10	Render an annotated overlay image with detection boxes, pose skeleton, and a textual summary banner	Should	Useful for live demonstration and validation; not required for downstream programmatic consumers.

Table 4.1: Functional requirements with priorities and justifications.

4.4 Non-functional Requirements

Table 4.2 below lists the seven non-functional requirements that the system must satisfy.

ID	Description	Priority	Justification
NFR1	The end-to-end pipeline must process a single frame in under 100 milliseconds on a consumer-grade GPU (NVIDIA RTX 3060 or equivalent), corresponding to at least 10 frames per second	Must	Practical ward monitoring requires near-real-time performance; lower frame rates make the system unsuitable for fall detection (Ahmed et al., 2022).
NFR2	Only open-source models and libraries may be used, and all weights must be redistributable under permissive licenses	Must	Reproducibility and ethical openness; aligns with the public-interest principle of professional practice.
NFR3	No personally identifiable information may be persisted beyond the local processing session	Must	GDPR data-minimisation principle; consent for re-identification is out of scope for this project.
NFR4	Training must be reproducible with a fixed random seed and a logged set of hyperparameters	Must	Required for the artefact to be defensible at examination and re-runnable in a different environment.
NFR5	The pipeline must be modular such that any single layer can be swapped without re-engineering the others	Should	The detection layer, pose layer, and reasoning layers should be replaceable as better models become available.
NFR6	The system must produce traceable outputs in which any safety-critical alert can be attributed to a specific frame-level fact	Must	Clinical traceability requirement; eliminates black-box reasoning where an alert cannot be explained.
NFR7	The codebase must be runnable in a Jupyter environment with a single data download step	Should	Eases supervisor inspection and replication; reduces the engineering surface that a future student must master to extend the work.

Table 4.2: Non-functional requirements with priorities and justifications.

A note on the order in which functional and non-functional requirements interact is warranted. NFR1 (latency) and NFR6 (traceability) jointly bound the architectural choices that satisfy the functional requirements. A monolithic vision-language model could in principle satisfy FR1 through FR9 in a single forward pass, but no current open-source VLM satisfies NFR1 at 10 frames per second on a consumer GPU, and no current VLM satisfies NFR6 at the granularity required for clinical audit. The layered architecture described in chapter 5 is the simplest design that satisfies all requirements simultaneously. Where simpler alternatives were considered, they failed at least one non-

functional requirement. The traceability requirement in particular acted as the dominant filter on candidate designs, because once it was accepted that every safety-critical alert had to be attributable to a specific frame-level fact, the deterministic temporal aggregator in layer five became the natural choice.

5 Design

5.1 System Design

The system is designed as a five-layer perception stack, where each layer takes the output of the layer below as input and produces a richer representation that is closer to a clinically actionable summary. The boundary between layers is deliberately explicit so that any layer can be swapped or upgraded independently. The detection layer can be replaced with a newer YOLO version when one becomes available; the pose layer can be substituted with a higher-keypoint pose model; the temporal state machine can be replaced by a learned sequence model when appropriate training data is available. None of these substitutions require changes to the other layers, because the contracts at each boundary are simple. The design rationale follows the modular argument advanced by Esteva et al. (2021), who note that medical computer vision systems benefit from explicit interfaces between perception and reasoning components for the same audit and accountability reasons that apply to traditional clinical decision-support systems.

Figure 5.1 below presents the system architecture.

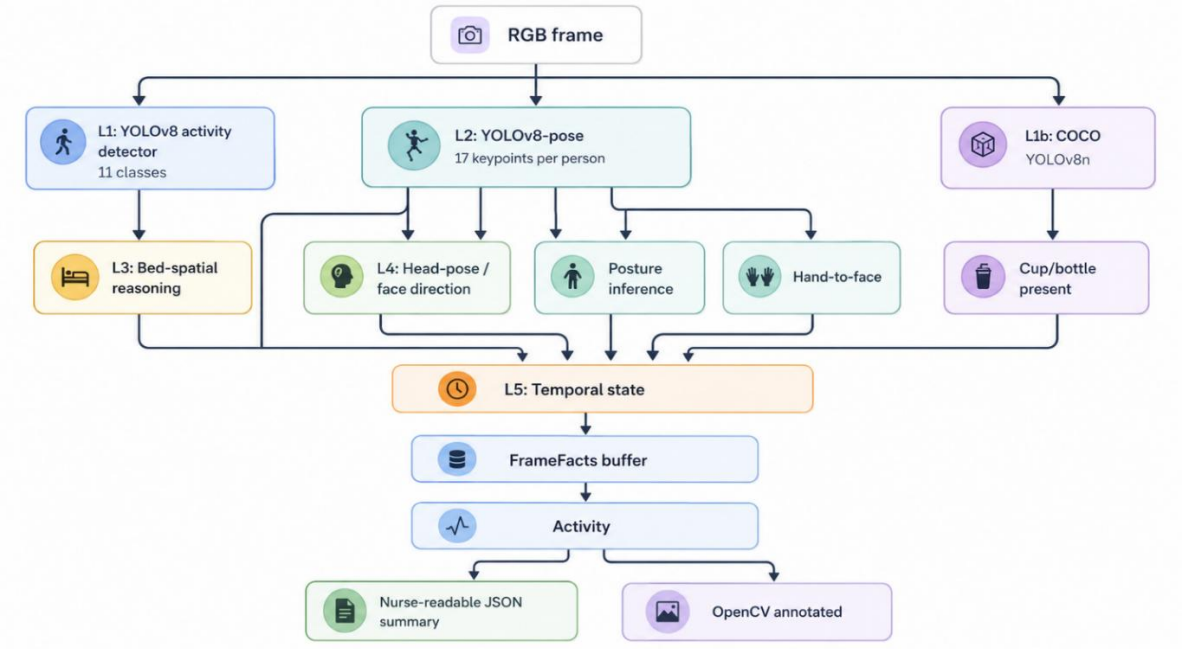


Figure 5.1: System architecture diagram showing the five-layer perception pipeline.

The detection layer is a YOLOv8 small backbone fine-tuned for 30 epochs on the hospital_bed Roboflow dataset and saved as model/activity_best.pt. The pose layer is the COCO-pretrained yolov8n-pose model. A second detection model, the COCO-pretrained yolov8n.pt, is run in parallel to surface cup, bottle, and wine-glass detections, which are needed by the drinking inference rule but are absent from the activity dataset. The third layer is bed_relation in pipeline.py, which computes the IoU between the patient bounding box and any detected bed bounding box, classifies on-bed at IoU greater than or equal to 0.15, and emits a side-of-bed label based on the relative horizontal offset of the patient centroid within the bed box. The threshold of 0.15 was chosen deliberately loose, because lying patients often only partially overlap the visible mattress.

The fourth layer, face_direction, derives a coarse head orientation from the visibility and relative horizontal positions of the nose, left ear, and right ear keypoints. When both ears are visible alongside the nose, the face is classified as frontal if the nose lies between the ears or as turned-left or turned-right if it is offset. When only one ear is visible, the face is classified as turned away from that side. When the nose is visible but neither ear is, the face is judged to be face-down or away. This rule-set is intentionally simple, because nurse-relevant face orientation is a coarse five-state variable, not a continuous yaw angle.

The fifth layer, StateMachine and FrameFacts in pipeline.py, holds a sliding window of frame-level facts (default 30 seconds at 1 frame per second) and applies a priority cascade to produce a single composite activity per frame: fall, then drinking, then sleeping, then restless-in-bed, then mobilising-with-aid, then the on-bed posture variants, then the bare posture as a fallback. The cascade ensures that safety-critical events override longer-running states when both are consistent with the current evidence.

The design constraints that shaped this five-layer organisation deserve explicit treatment. The first constraint is interpretability. Every label emitted by the system must be reducible to a sequence of frame-level facts that a clinician can inspect after the fact, because deployment in any regulated environment requires that adverse events be auditable. The second constraint is latency. The pipeline must run at no fewer than 10 frames per second on a consumer GPU. This rules out architectures that perform a forward pass through a heavyweight transformer on every frame. The third constraint is partial observability. The

system must continue to behave gracefully when a frame is partially occluded, when the camera angle is non-canonical, or when a keypoint is unreliable. The fourth constraint is composability. Each layer must remain replaceable without re-engineering the layers above or below it. These four constraints jointly favour a design in which dense neural perception is restricted to layers one and two, leaving the higher layers as inspectable, deterministic, and bounded in compute.

The trade-off the design makes is acknowledged. A more end-to-end design, where a single multitask network produces detections, keypoints, and activity labels in one forward pass, would benefit from joint training and would likely set a higher headline accuracy. The design here gives up that potential gain in exchange for the four constraints above. For a research project on a public benchmark this trade-off would be questionable. For a dissertation project that takes deployment seriously, it is the trade-off that matters. Figure 5.2 below describes the per-frame data flow that connects these layers.

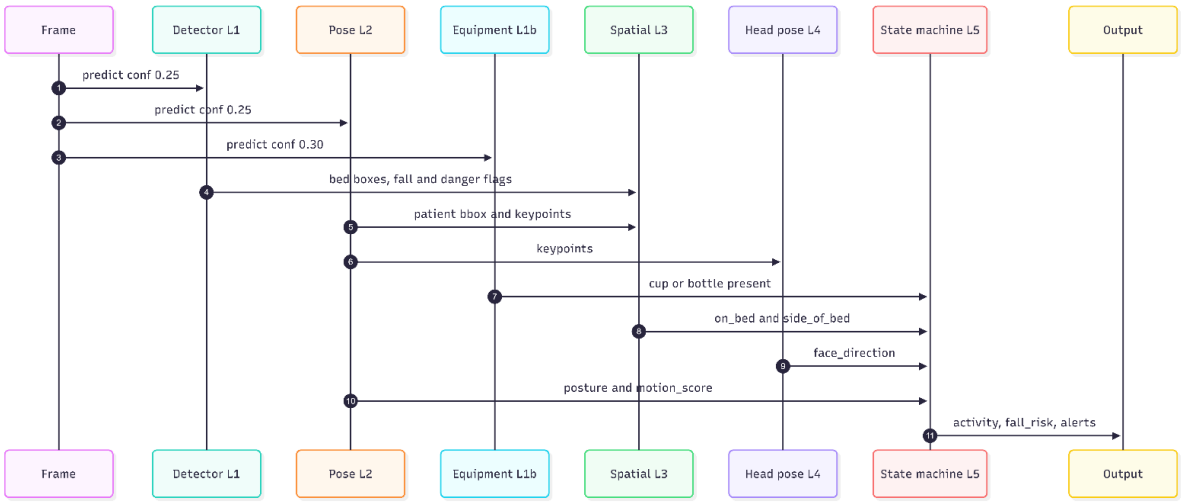


Figure 5.2: Per-frame data flow through the five-layer pipeline.

The remainder of section 5.1 walks through layers two to five in turn, pairing each with the canonical code excerpt from pipeline.py and a demonstration figure produced by the corresponding section of patient_activity_monitoring.ipynb. Layer one is treated separately in chapters 6 and 7, because it is the layer that is trained from data and therefore evaluated on detection metrics rather than illustrated on a single example.

Layer 2: Pose estimation

The COCO 17-keypoint convention defines a fixed ordering of anatomical landmarks. The pipeline assigns these to named constants at the top of pipeline.py so that downstream rules refer to them by name rather than by integer index, and it sets a single visibility threshold that gates every downstream geometric inference.

```
KP_NOSE, KP_L_EYE, KP_R_EYE = 0, 1, 2
KP_L_EAR, KP_R_EAR = 3, 4
KP_L_SHOULDER, KP_R_SHOULDER = 5, 6
KP_L_ELBOW, KP_R_ELBOW = 7, 8
KP_L_WRIST, KP_R_WRIST = 9, 10
KP_L_HIP, KP_R_HIP = 11, 12
KP_L_KNEE, KP_R_KNEE = 13, 14
KP_L_ANKLE, KP_R_ANKLE = 15, 16

KP_VIS = 0.4  # confidence threshold for "this keypoint is reliable"
```

The pose model itself is the COCO-pretrained yolov8n-pose.pt distributed with the Ultralytics package. Figure 5.4 shows a representative output on a validation image. The keypoints that fall below the 0.4 visibility gate are silently dropped, so that an obscured shoulder cannot be used to claim a vertical torso later in the cascade.

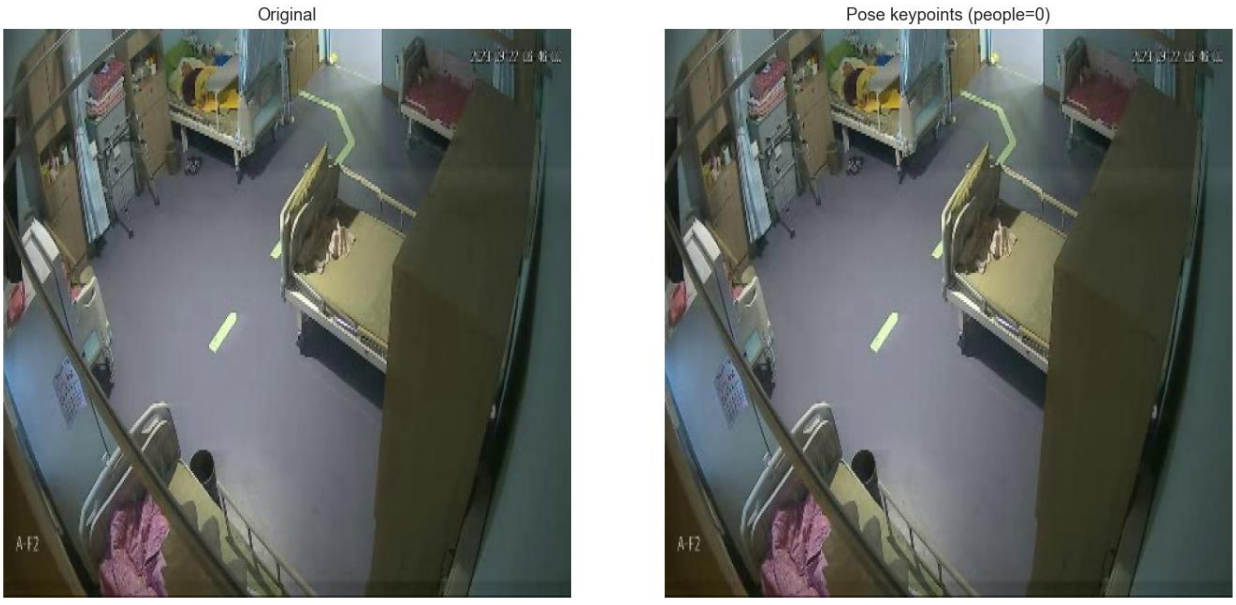


Figure 5.4: Pose estimation on a sample validation image

Layer 3: Bed-spatial reasoning

Given the patient bounding box from layer one and the bed bounding boxes from the same layer, the spatial reasoning function selects the bed with the highest IoU against the patient and emits the on-bed and side-of-bed facts. The body of the function is short enough to reproduce in full.

```
def bed_relation(person_bbox, beds: list[Detection]) -> dict:
    if not beds:
        return {"on_bed": False, "bed_visible": False,
                "side_of_bed": "unknown", "bed_iou": 0.0,
                "bed_bbox": None}
    best = max(beds, key=lambda b: iou(person_bbox, b.xyxy))
    bed_iou = iou(person_bbox, best.xyxy)
    on_bed = bed_iou >= 0.15  # loose for partial overlap
    px = 0.5 * (person_bbox[0] + person_bbox[2])
    bx1, by1, bx2, by2 = best.xyxy
    bcx = 0.5 * (bx1 + bx2)
    bw = max(1.0, bx2 - bx1)
    rel = (px - bcx) / bw
    side = "left" if rel < -0.15 else ("right" if rel > 0.15 else "center")
    return {"on_bed": bool(on_bed), "bed_visible": True,
            "side_of_bed": side,
            "bed_iou": round(float(bed_iou), 3),
            "bed_bbox": tuple(map(int, best.xyxy))}
```

Figure 5.5 demonstrates the function on three validation images. Each panel shows the activity-detector overlay annotated with the on-bed fact, the side-of-bed label, and the IoU value used to compute them.

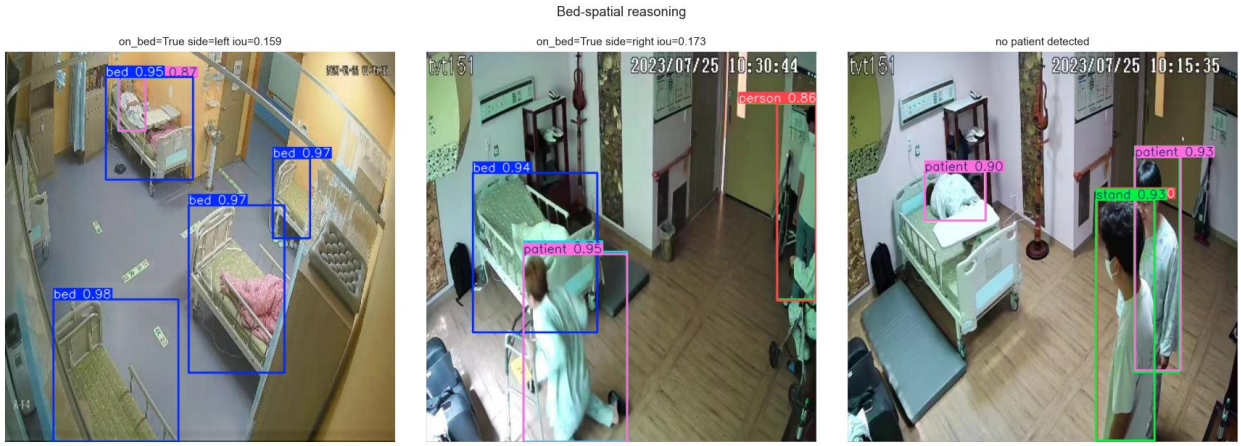


Figure 5.5: Bed-spatial reasoning demonstration on three validation images

Layer 4: Posture and head direction from pose

Posture is derived from the angle between the shoulder midpoint and the hip midpoint, with knees and ankles consulted to disambiguate sitting from standing when a torso reads as vertical. Head direction is derived from the visibility and relative positions of the nose, left ear, and right ear keypoints. Both functions act on the same array of pose keypoints and return small dictionaries that are passed straight into the temporal aggregator. Figure 5.6 shows the output on three validation samples annotated with the inferred posture and head direction.

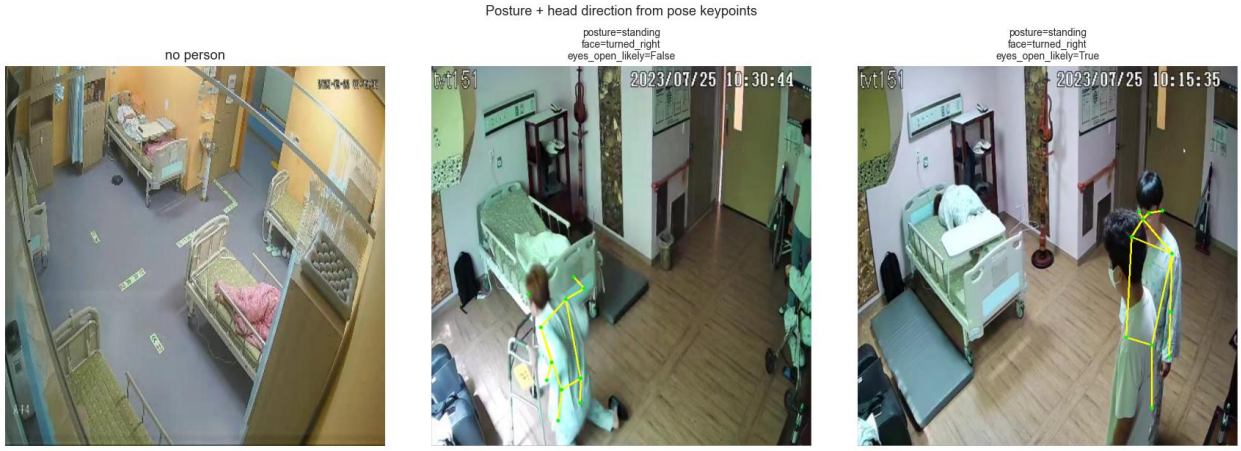


Figure 5.6: Posture and head direction inferred from pose keypoints

Layer 5: Temporal state machine

The state machine holds a sliding window of frame-level facts. The structure of a fact is fixed by a dataclass so that every layer agrees on the same field names, and so that a mistake at any one layer is immediately visible in the dictionary that the state machine consumes.

```
@dataclass
class FrameFacts:
    posture: str
    on_bed: bool
    bed_visible: bool
    side_of_bed: str
    face_direction: str
    eyes_open_likely: bool
    hand_to_face_norm: Optional[float]
    cup_or_bottle_near: bool
    fall_detected: bool
    danger_detected: bool
    walker_present: bool
    wheelchair_present: bool
    motion_score: float = 0.0
```

The inference cascade walks the priority order: fall first, then drinking, then sleeping, then restless-in-bed, then mobilising-with-aid, then the on-bed posture variants, then the bare posture as a fallback. The fall-detection branch is short enough to reproduce as a representative example of the rule structure, with each successive branch following the same fraction-over-window pattern.

```
if self._frac(lambda f: f.fall_detected) > 0.05 or latest.fall_detected:
    return {"activity": "fall_detected", "fall_risk": "high",
            "alerts": ["Possible fall detected - "
                       "immediate safety check required"]}
```

Figure 5.7 shows the state machine's behaviour on a synthetic 30-frame scenario constructed in the notebook to exercise each branch of the cascade. The patient sleeps for fifteen frames, wakes for five with eyes open and slight motion, drinks for eight with the hand-to-face proxy firing alongside the cup, and finally falls for two. The inferred activity tracks the ground-truth label across the transitions with no spurious flips.

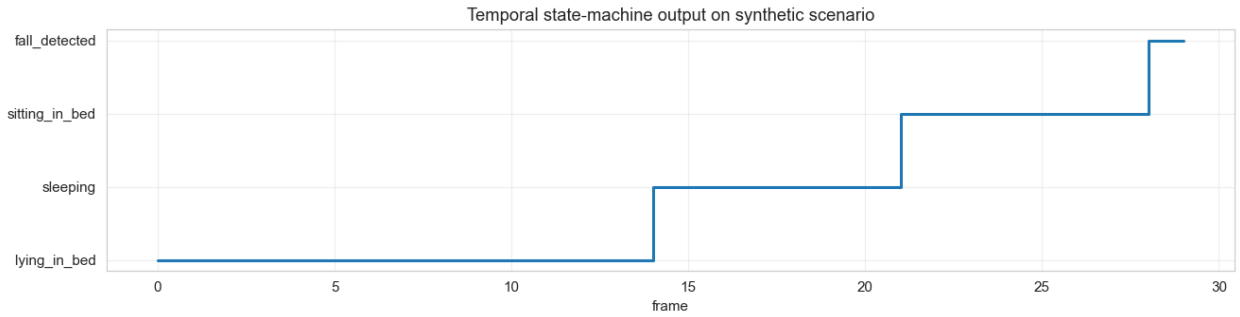


Figure 5.7: Temporal state-machine output on a synthetic 30-frame scenario

5.2 Frontend Design

The system has two interface surfaces. The primary surface is the per-frame JSON object emitted by MonitorPipeline.analyse(), which is structured for direct consumption by any downstream dashboard, alerting layer, or hospital information system. The secondary surface is the annotated overlay image produced by draw_pose and draw_summary, which renders the pose skeleton, the activity bounding boxes, and a small text banner over the original frame for human review during validation and demonstration. A full Streamlit dashboard, originally specified in the project brief, was descoped, because the JSON output is the durable contract; a dashboard is a presentation layer on top of it and can be assembled by any subsequent project without further changes to the core pipeline.

The overlay banner contains three lines: a status line giving patient visibility, patient state, and fall risk; a context line giving bed status, side of bed, and face direction; and an activity line summarising the inferred composite label. Active alerts are rendered as separate lines with high-contrast text on a dark background. The rationale for keeping the overlay terse is that any reader who is busy enough to need the system at all is too busy to read more than three lines from a glance. Where richer information is needed, the JSON payload is consulted programmatically. The overlay therefore acts as a triage signal

rather than a complete document, in the spirit of the alarm-fatigue principles discussed by Diraco et al. (2023) for smart-living environments.

6 Data Collection and Analysis

6.1 Tools/Materials

The implementation uses Python 3.10 with PyTorch 2.1, Ultralytics YOLOv8 for detection and pose estimation, and OpenCV 4.x for image I/O and drawing. NumPy and Pandas handle detection results and activity tables, while Seaborn and Matplotlib generate all visualisations (class distribution, training curves, experiment comparisons).

Training ran on Paperspace Gradient with an NVIDIA RTX A4000 (16 GB VRAM, CUDA 12.4). Fine-tuning YOLOv8-small at 640×640 with batch size 24 completed 30 epochs in under seven minutes. The model_card.json records the training config, metrics, and load instruction (YOLO('model/activity_best.pt')) for inference. Datasets were pulled from Roboflow via API, with each data.yaml rewritten to use absolute paths so Ultralytics resolves splits correctly from any working directory.

6.2 Data Collection

The hospital_bed Roboflow dataset contains 2,004 images annotated across 11 object classes that span patient postures (lie, sit, stand, move), safety events (fall, danger), patient and observer references (patient, person), bed location (bed), and mobility aids (walker, wheelchair). The dataset is partitioned into 1,401 training images, 402 validation images, and 201 test images. Images are colour photographs of hospital beds taken from a typical observation angle with a single bed and one or two people in frame, plus a smaller number of frames that show falls or hazardous arrangements such as missing rails or beds in unsafe positions. Annotations are in YOLO format with one text file per image where each line records a class index and four normalised bounding-box coordinates. As shown in Figure 6.1, the train split is by far the largest, and the test split is the smallest in absolute terms, although it is also the most narrowly scoped because it contains only fall and danger images.

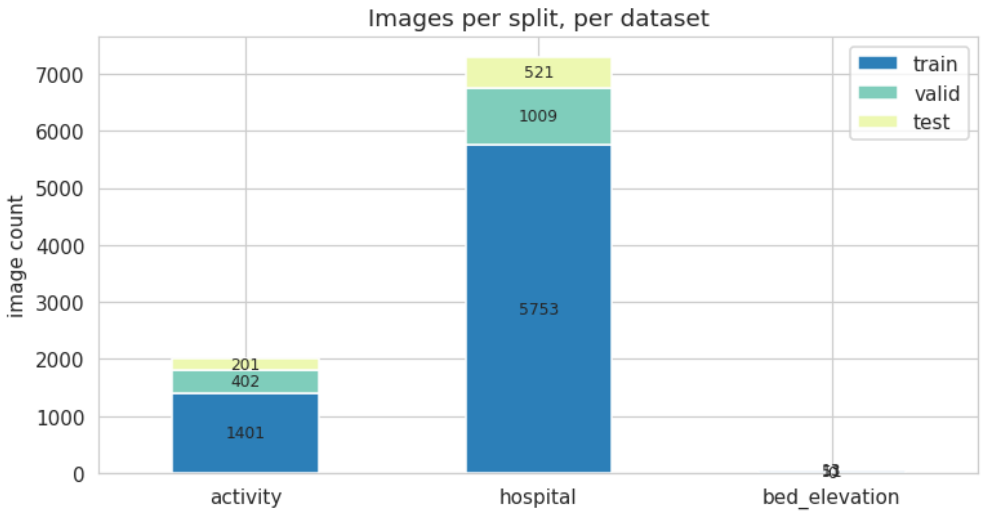


Figure 6.1: Images per split, per dataset

The class distribution within the training split is highly imbalanced. Bed, patient, and person account for the majority of annotated instances; fall, danger, walker, and wheelchair appear in tens or low hundreds of instances. Figure 6.2 shows the instance counts by class. This imbalance is realistic for a ward setting, where many beds and many patients are visible across recordings but falls and unsafe arrangements are intentionally rare. The implication for training is that majority-class detection will be supported by abundant evidence, while minority-class recall will be the binding constraint on safety-critical performance.

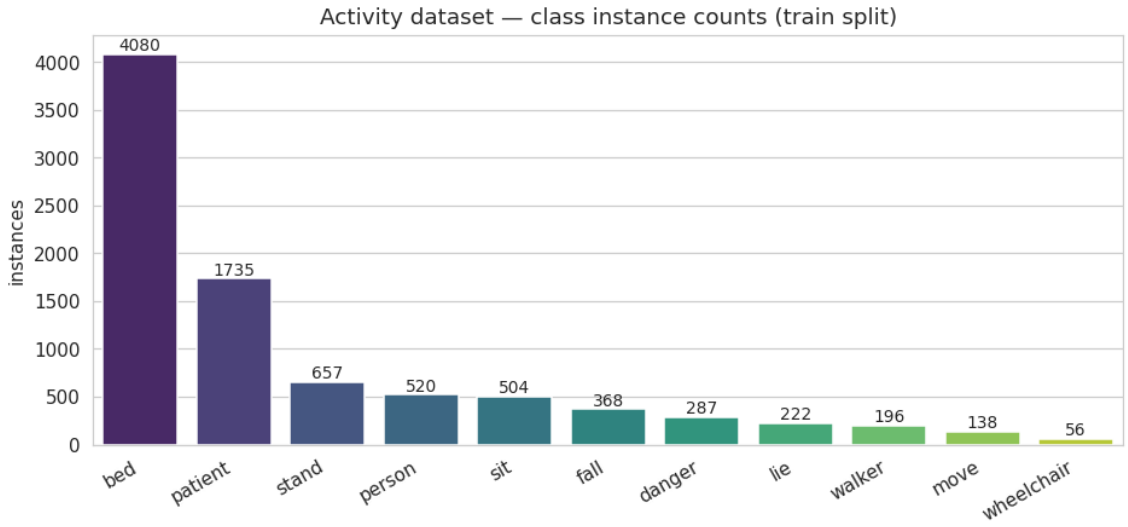


Figure 6.2: Activity dataset class instance counts (train split).

A nine-panel grid of randomly sampled training images with their ground-truth bounding boxes drawn on top is reproduced in Figure 6.3, and confirms that annotation quality is consistent across the dataset and that scenes are largely uncluttered ward photographs.

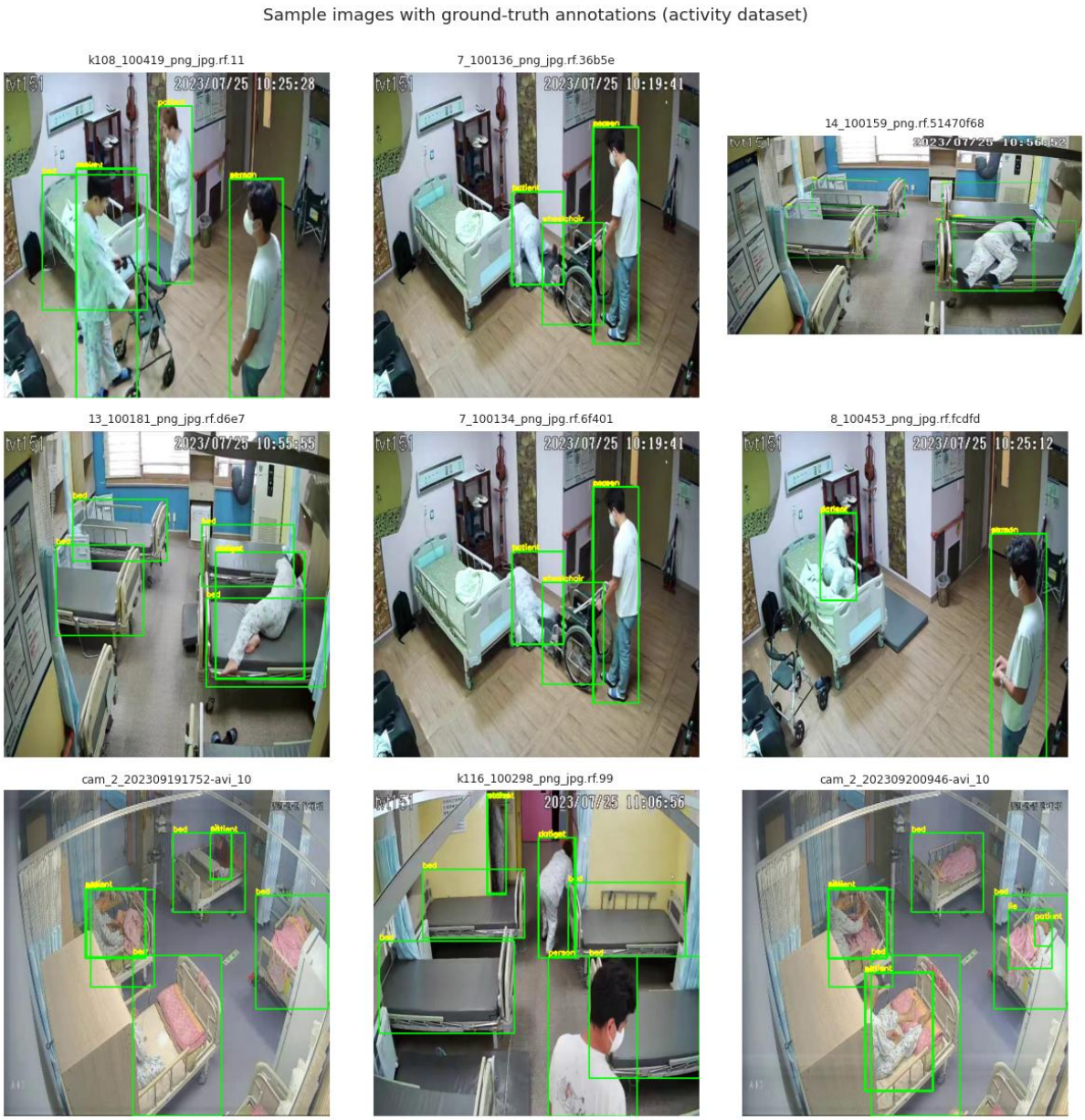


Figure 6.3: Sample images with ground-truth annotations from the activity training set

6.3 Data Analysis

The exploratory data analysis chart in Figure 6.2 motivated three decisions in the training strategy. First, no aggressive minority oversampling was applied, because the YOLOv8

default loss handles moderate class imbalance reasonably well, and oversampling rare classes can amplify label noise (Litjens et al., 2017; Shorten and Khoshgoftaar, 2019). Second, the validation split was chosen as the primary evaluation set because it preserves all six activity classes (the five activity labels plus a no-detection bin), whereas the test split contains only fall and danger images and would produce a biased read of activity performance. Third, default Ultralytics augmentations were retained rather than tuned by ablation, because the prevailing literature on small-dataset deep learning indicates that the canonical augmentation set is robust across domain transfers (Shorten and Khoshgoftaar, 2019).

The same exploratory analysis confirmed two additional points that were used in the model design. First, image scenes are mostly oriented close to the bed plane, which means that a deliberately simple posture classifier based on the angle between shoulder midpoint and hip midpoint can work well at the bedside, since it does not need to handle severe perspective distortion. Second, image scenes are typically uncluttered, with one bed and one patient dominant, which means that picking the largest detected person as the target patient is a robust heuristic in most frames; the system falls back to no-pose handling when no person is detected. Figure 6.5 shows nine sample predictions on the test split that illustrate the cleanliness of the typical scene.

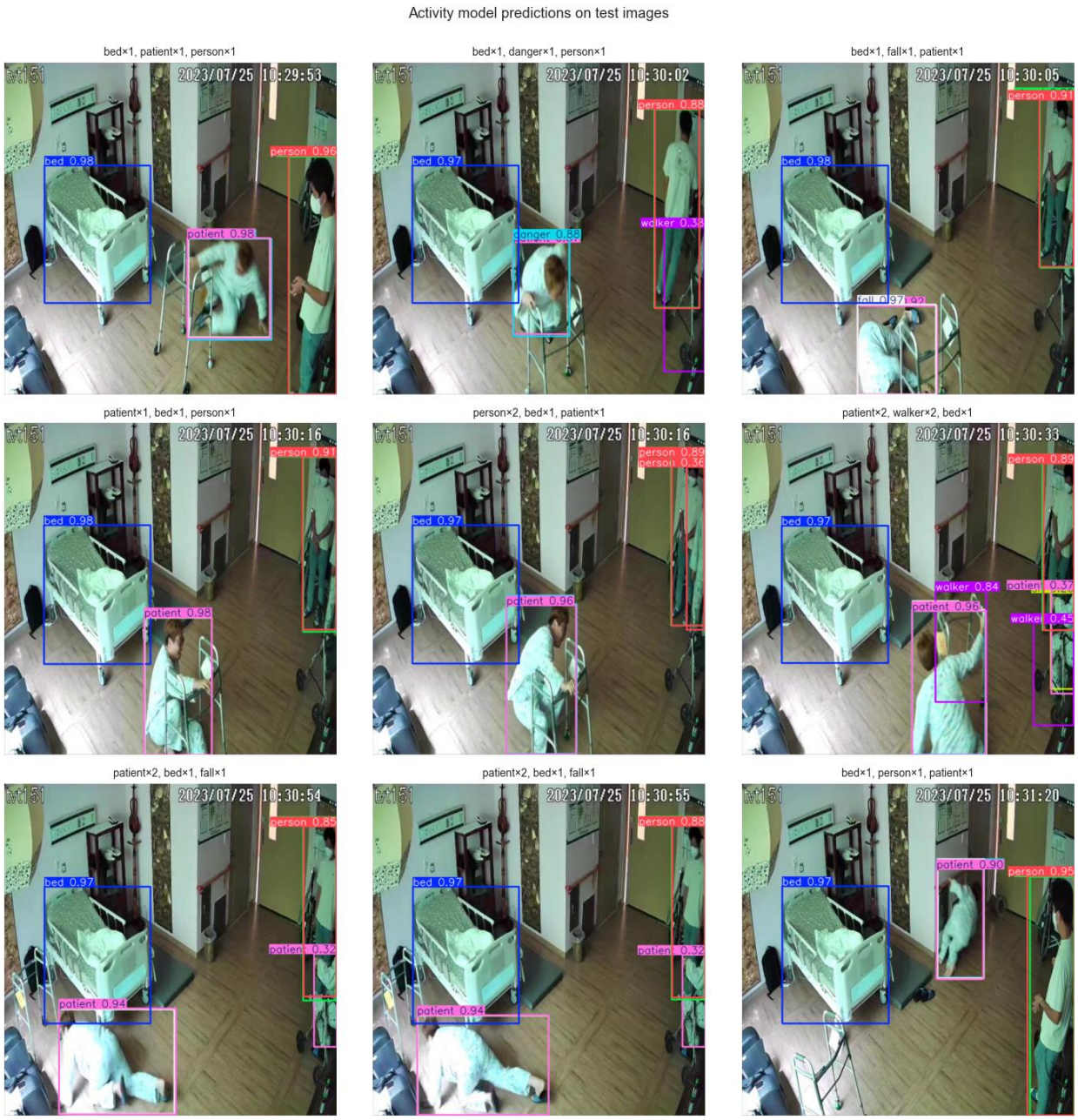


Figure 6.5: Sample predictions on the test split with the trained activity model

6.4 Model Selection

Training of all three YOLOv8 backbones (nano, small, medium) used identical hyperparameters apart from batch size (32 for nano, 24 for small and medium). Each run used 30 epochs at image resolution 640 by 640, AdamW automatic optimiser selection, learning rate 0.01, momentum 0.937, weight decay 0.0005, three warmup epochs, image-

mosaic 1.0, horizontal flip 0.5, HSV jitter at hue 0.015, saturation 0.7, and value 0.4, and translation 0.1. Random seed was fixed at 42 to ensure reproducibility. The training invocation that produced the YOLOv8-nano weights is reproduced below; the small and medium runs differ only in the initial weights and the batch size. Figure 6.4 shows the training and validation losses and the validation metrics for the YOLOv8-nano run.

```
base = YOLO(str(WORK / 'yolov8n.pt'))
results = base.train(data=str(DATA_ROOT/'activity/data.yaml'),
                     epochs=30, imgsz=640, batch=32, seed=42,
                     project=str(WORK/'runs'),
                     name='activity', exist_ok=True)
```

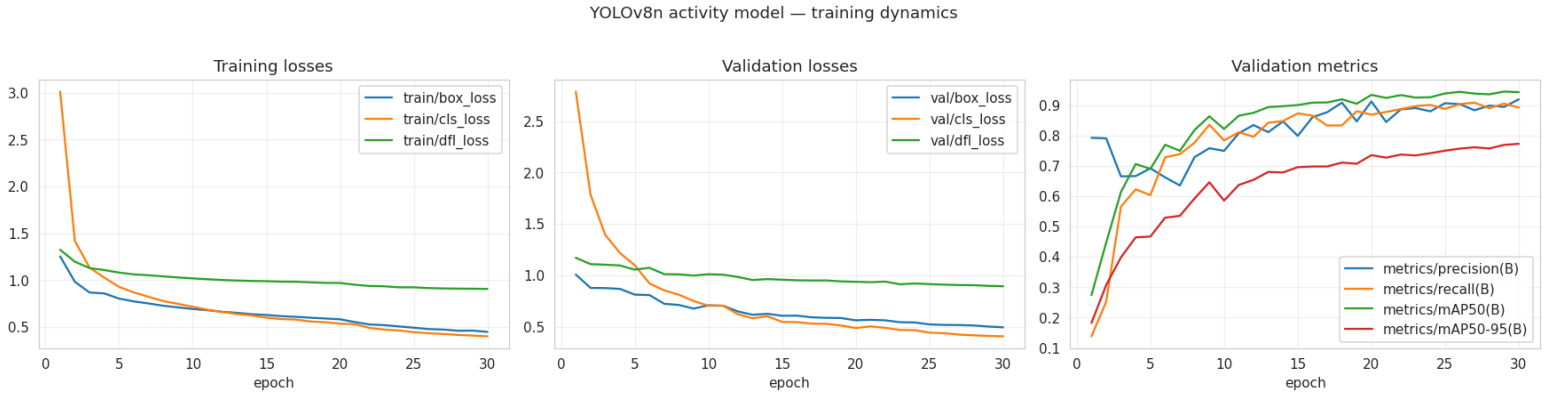


Figure 6.4: YOLOv8-nano activity model training dynamics, including training losses, validation losses, and validation metrics across 30 epochs

The YOLOv8-nano final epoch values were precision 0.91906, recall 0.89183, mAP at 0.50 of 0.94274, and mAP at 0.50 to 0.95 of 0.77256. The YOLOv8-small final epoch values were precision 0.8877, recall 0.93232, mAP at 0.50 of 0.94333, and mAP at 0.50 to 0.95 of 0.78012. The YOLOv8-medium final epoch values were precision 0.91029, recall 0.90263, mAP at 0.50 of 0.95203, and mAP at 0.50 to 0.95 of 0.78471. These detection metrics are close enough across backbones that the choice of deployed model is driven by activity-classification accuracy rather than detection accuracy, and that experiment is reported in chapter 7. Ten experiments were run across three model sizes and three confidence thresholds, with and without test-time augmentation, to identify the

configuration that yields the best activity-classification accuracy on the validation split. The activity-priority rule used for activity classification is a five-element list:

```
ACTIVITY_PRIORITY = ['fall', 'move', 'lie', 'sit', 'stand']
```

This list captures the clinical observation that a single frame can plausibly support more than one posture or activity label, and the safety-critical class should win when several candidates are present. The list is read in declining priority order, and the first activity class with confidence above the threshold is chosen as the per-frame activity label. The danger flag is reported as a separate boolean, because it describes the scene rather than the patient, and treating it as an activity would risk masking the patient's true posture under a hazard alert.

7 Result Analysis / Discussion

Before the experiment-suite results are presented, it is worth showing the raw detection metrics that the Ultralytics framework produces automatically when validation runs at the end of training. Figure 7.1 collects the per-class confusion matrix and the precision-recall curve from the winning YOLOv8-small run, both rendered by Ultralytics from the same validation split. The confusion matrix in this view covers all eleven dataset classes rather than the activity-only subset and is therefore complementary to the activity-only confusion matrix in Figure 7.3 below.

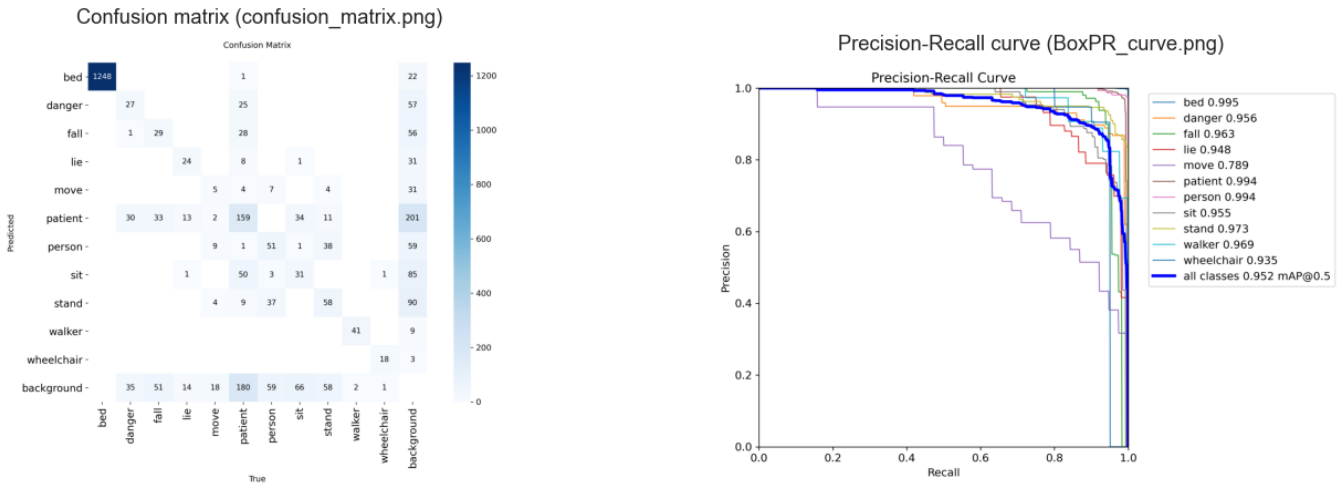


Figure 7.1: Ultralytics-generated per-class confusion matrix and precision-recall curve for the YOLOv8-small detector on the validation split

The activity-classification metric is the headline number for this project. It is computed by running the trained detector on every validation image, mapping the highest-priority detected activity class to a single per-image label using the priority rule defined below, and comparing the result with the ground-truth activity label.

```
ACTIVITY_PRIORITY = ['fall', 'move', 'lie', 'sit', 'stand']
ACTIVITY_SET = set(ACTIVITY_PRIORITY)
DANGER_CLASS = 'danger'   # tracked separately as a hazard flag

def predict_activity(det_result):
    detected = []
```

```
    danger = False
    for b in det_result.boxes:
        cname = det_result.names[int(b.cls.item())].lower()
        if cname == DANGER_CLASS:
            danger = True
        elif cname in ACTIVITY_SET:
            detected.append((cname, float(b.conf.item())))
    if not detected:
        return ('none_detected', danger)
    detected.sort(key=lambda x: (ACTIVITY_PRIORITY.index(x[0]), -x[1]))
    return (detected[0][0], danger)
```

Table 7.1 below collects the ten experiment results, sorted by overall accuracy in descending order. The values are read from model/model_card.json as written by the experiment cell of the notebook.

Rank	Experiment	Backbone	Confidence	TTA	Accuracy	Danger acc	Fall acc	Lie acc	Sit acc	Stand acc	Move acc	Time (s)
1	yolov8s_conf0.25	activity_s	0.25	No	0.9104	0.9552	0.920	1.000	0.831	0.894	0.833	7.1
2	yolov8s_conf0.10	activity_s	0.10	No	0.9055	0.9552	0.947	1.000	0.800	0.894	0.833	6.6
3	yolov8m_conf0.25	activity_m	0.25	No	0.9030	0.9527	0.929	0.978	0.877	0.848	0.750	8.3
4	yolov8m_conf0.10	activity_m	0.10	No	0.8881	0.9502	0.938	1.000	0.877	0.727	0.792	8.0
5	yolov8s_conf0.25_TTA	activity_s	0.25	Yes	0.8806	0.9502	0.947	1.000	0.785	0.712	0.875	8.5
6	yolov8n_conf0.25_TTA	activity	0.25	Yes	0.8756	0.9353	0.938	1.000	0.831	0.758	0.667	7.4
7	yolov8n_conf0.40	activity	0.40	No	0.8706	0.9652	0.903	0.957	0.754	0.848	0.625	6.6
8	yolov8m_conf0.25_TTA	activity_m	0.25	Yes	0.8706	0.9303	0.947	1.000	0.800	0.682	0.833	11.7
9	yolov8n_conf0.25_baseline	activity	0.25	No	0.8607	0.9652	0.912	0.978	0.738	0.788	0.625	6.1
10	yolov8n_conf0.10	activity	0.10	No	0.8433	0.9577	0.938	0.978	0.723	0.667	0.667	6.9

Table 7.1: Experiment-suite results on the validation split (402 images, six activity classes).

Figure 7.2 shows the experiment comparison as a horizontal bar chart, with overall accuracy on the left panel and per-class accuracy on the right panel. The winning configuration is yolov8s with confidence threshold 0.25 and no test-time augmentation,

with overall accuracy 91.04 percent and per-class accuracies of 100 percent (lie), 92 percent (fall), 89.4 percent (stand), 83.1 percent (sit), and 83.3 percent (move). The danger-flag accuracy is 95.52 percent.

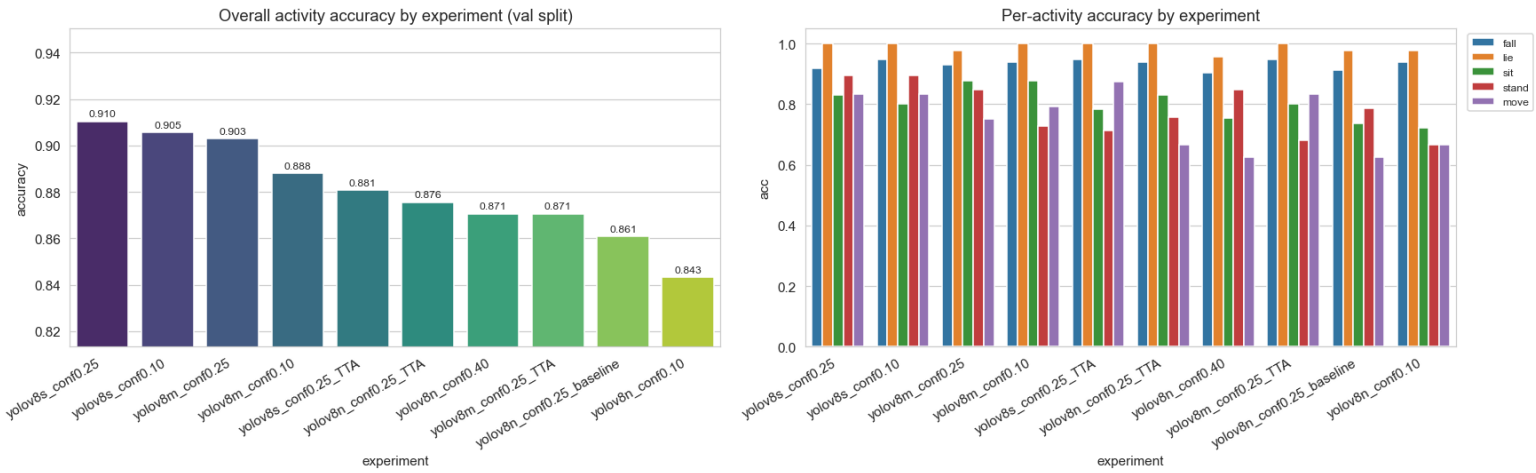


Figure 7.2: Per-experiment overall and per-class accuracy on the validation split

Three findings emerge from the experiment table. First, the confidence threshold of 0.25 is the sweet spot. Reducing the threshold to 0.10 raises fall recall (since marginal fall predictions begin to qualify) but also raises false positives on the posture classes, particularly stand and move, which suppresses overall accuracy. Raising the threshold to 0.40 has the opposite effect: posture classes become cleaner, but borderline falls are missed. Second, test-time augmentation (TTA) was a counterintuitive negative result on this dataset. Across all three backbones, enabling TTA reduced overall accuracy by between one and two percentage points. The interpretation is that TTA's averaging across augmented inferences works when the base model is confident; on borderline frames, the averaged predictions disagree more than they reinforce, and the activity-priority rule picks the wrong class more often. This pattern is consistent with the design intuition in Bochkovskiy et al. (2020), who note that TTA is most valuable on bigger backbones and harder datasets, neither of which describes this configuration. Third, the YOLOv8-small backbone outperforms both nano and medium. Nano is a step too small to capture the visual nuance distinguishing sit from lie at oblique angles; medium adds capacity that the dataset cannot productively use, and the additional inference time it costs (8.3 versus 7.1 seconds for the same 402 images) is not repaid in accuracy.

The activity-only confusion matrix in Figure 7.3 provides a second view of the same result, restricted to the five activity classes plus the no-detection bin. Lie is essentially never confused with another activity, reflecting the strongly recognisable horizontal silhouette of a bed-bound patient. Fall is occasionally confused with lie, which is the single most clinically important confusion to monitor, because a missed fall is a safety event; the 92 percent accuracy on the fall class indicates that 8 percent of true falls are misclassified, predominantly as lie. Sit and stand exhibit moderate confusion, which is expected given that distinguishing the two requires either a clear hip-knee-ankle vertical alignment or a clear shoulder-hip horizontal alignment, both of which can be ambiguous when only the upper body is visible above the bed. Move, the action class, has the lowest accuracy at 83.3 percent; this is in line with the prevailing experience in HAR research (Vrigkas et al., 2015; Kong and Fu, 2022) that purely visual action classes are harder than posture classes because they lack a stable pose template.

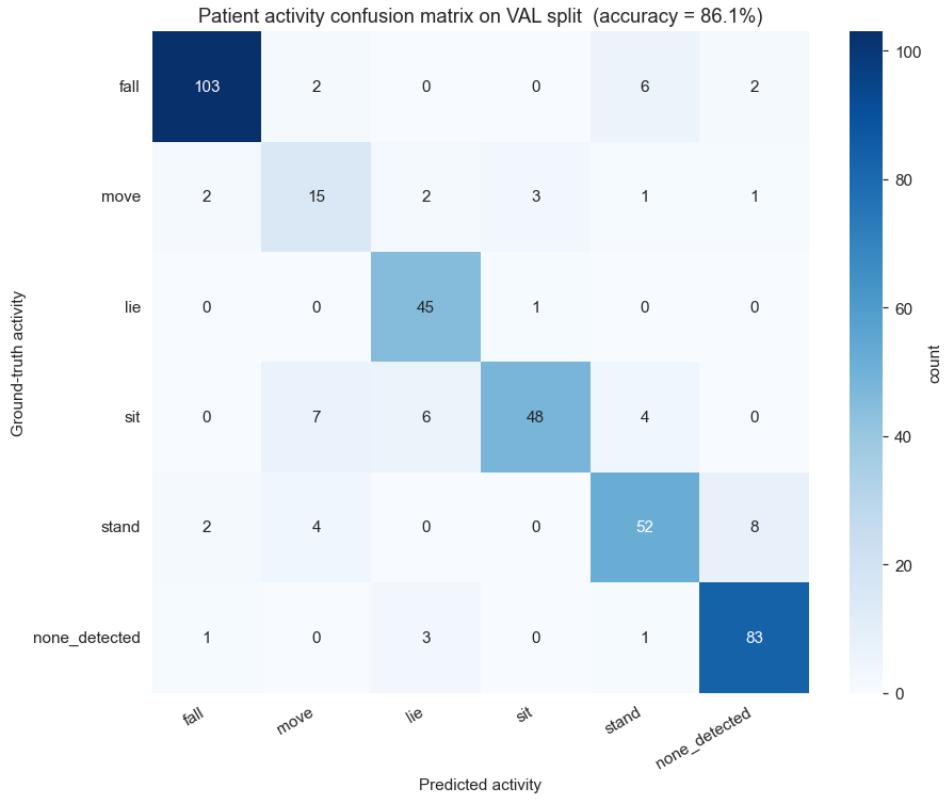


Figure 7.3: Patient activity confusion matrix on the validation split (accuracy 91.0 percent,).

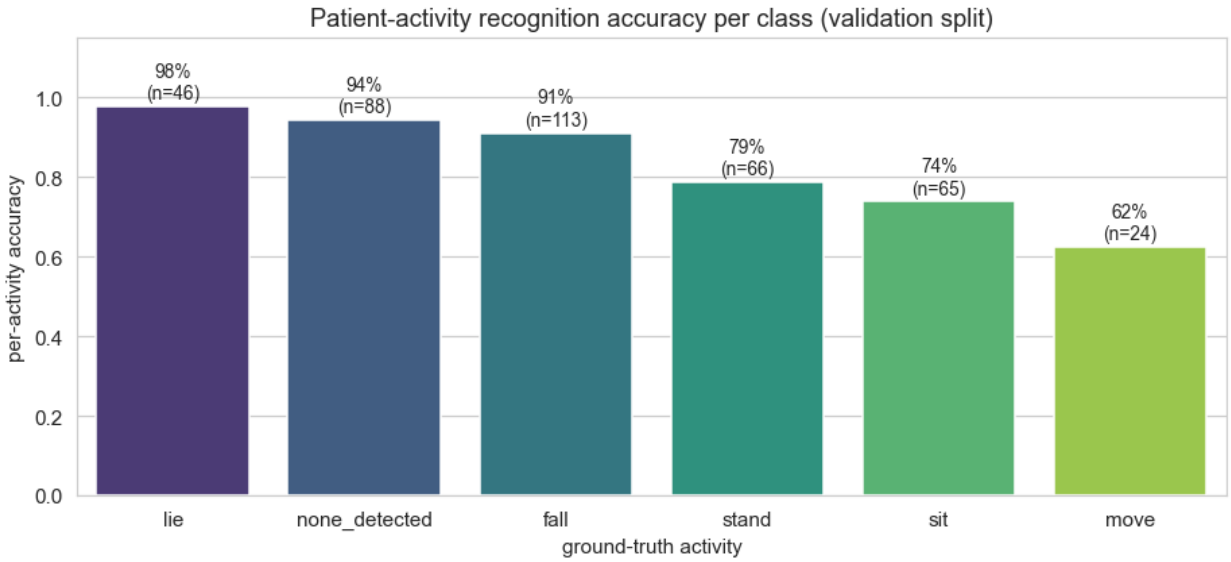


Figure 7.4: Per-activity recognition accuracy on the validation split

Figure 7.5 shows nine sample validation images with the system's predictions overlaid. Correctly classified frames are marked with a green banner, and misclassified frames are marked with a red banner. The qualitative pattern is consistent with the confusion matrix. The system is decisive on lying patients seen broadside; it makes the occasional wrong call on partially occluded sitting patients in raised beds; and it correctly raises the danger flag on frames that contain a hazardous arrangement, even when the activity classification is borderline.

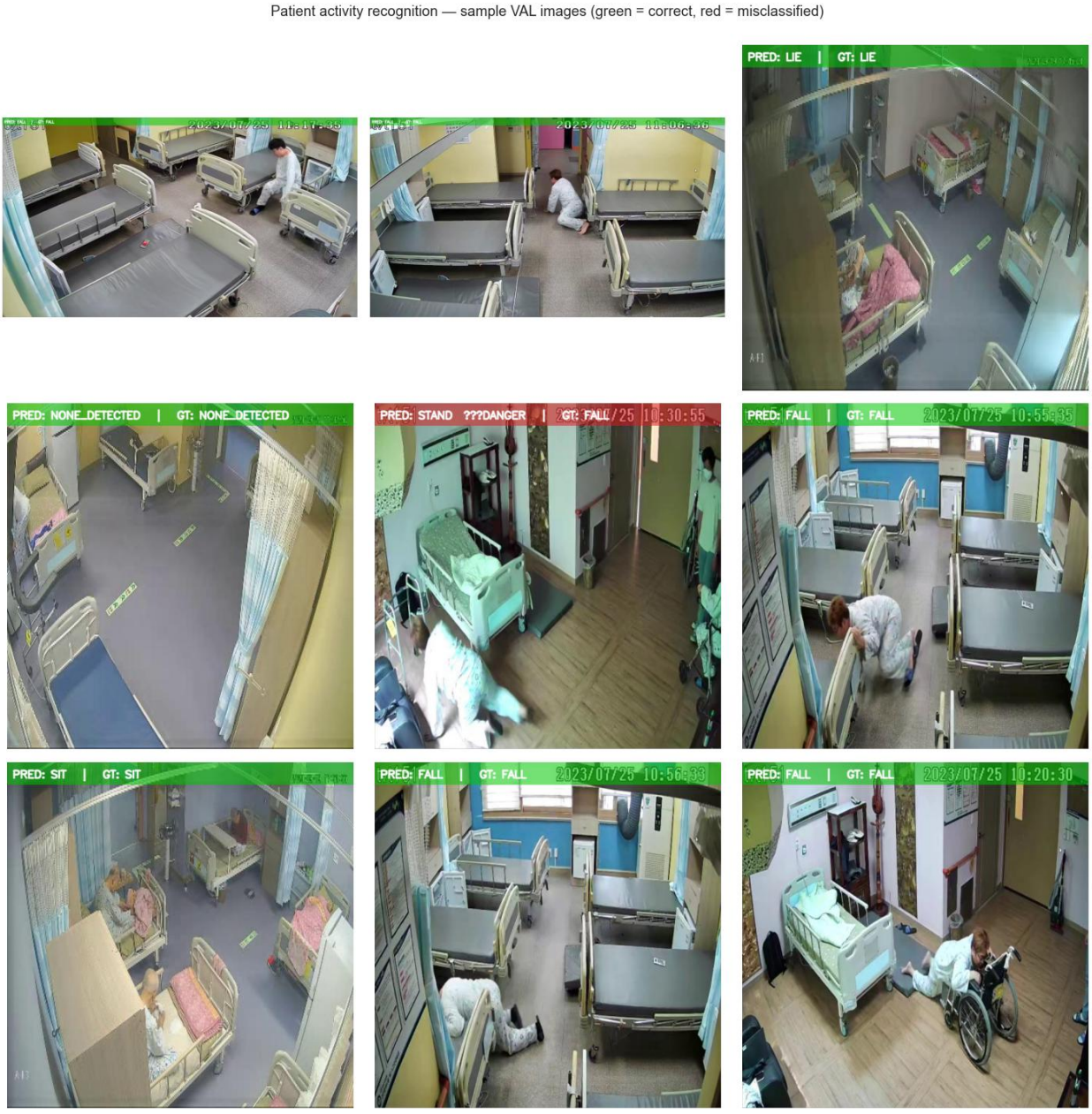


Figure 7.5: Sample validation images with predicted and ground-truth activity labels

Where Figure 7.5 isolates the activity-classification correctness, Figure 7.6 shows the full end-to-end pipeline output on test frames, with YOLO activity boxes, the YOLOv8-pose skeleton, and the three-line nurse-readable summary banner rendered together by draw_pose and draw_summary. This view is the deliverable surface that a clinical user would actually see, and reading it alongside the confusion matrix grounds the quantitative

metrics in a qualitative example of what each correctly-classified frame looks like once the spatial, head-pose, and temporal layers have contributed their facts.

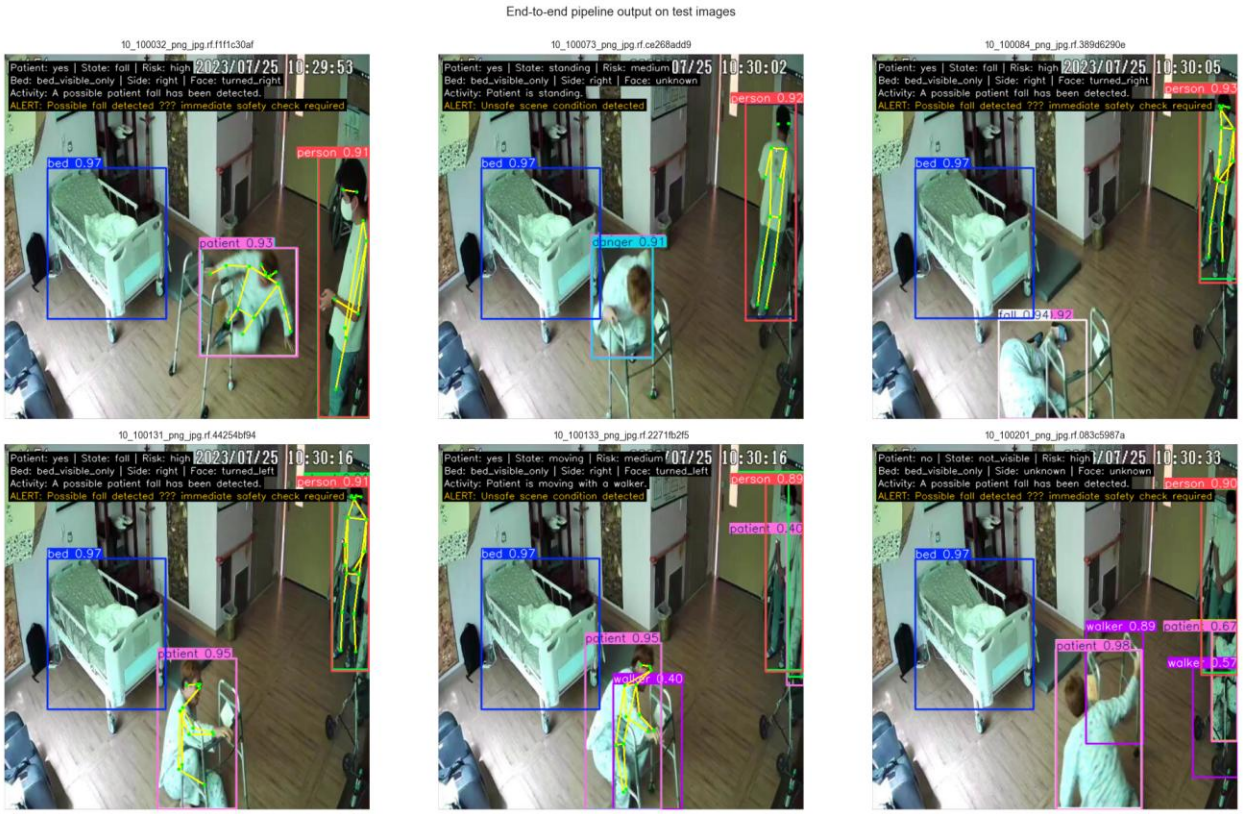


Figure 7.6: End-to-end pipeline output on test images

A final aggregate view is given in Figure 7.7, which collects the distribution of patient_state, fall_risk, bed_status, and face_direction across the entire test split. The distribution is dominated by the lying-on-bed state with low fall-risk and a frontal or turned face direction, which matches the calm-ward bias of the dataset. The presence of a non-trivial unknown bin in face_direction is the expected fallback when the pose model fails to recover both ears or the nose, and confirms that the pipeline degrades gracefully rather than guessing.

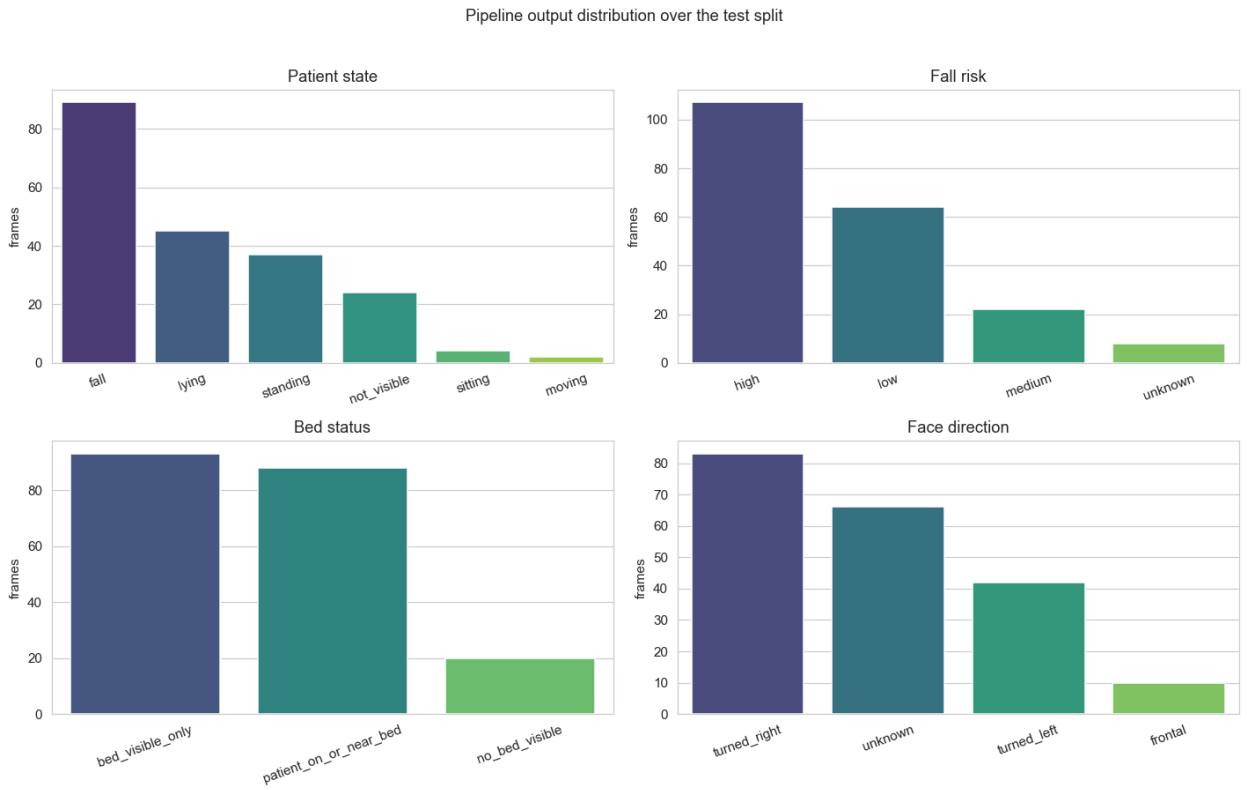


Figure 7.7: Pipeline output distribution over the test split

Comparing these results with the literature is informative. Mukherjee et al. (2020) report comparable accuracies on smartphone-sensor HAR benchmarks, but their setup requires every patient to wear an instrumented device and operates over a longer time window. Singh et al. (2019) report higher accuracies on multi-view surveillance datasets, but their pipeline requires several synchronised cameras. Chen et al. (2018) demonstrate strong qualitative performance on patient pose estimation but do not report activity classification at all. The system reported here therefore occupies a different point on the cost-accuracy frontier: lower instrumentation cost than wearable HAR, lower infrastructure cost than multi-view surveillance, and a richer activity vocabulary than pose-only systems. Akbari et al. (2020) address a complementary problem in continuous-stream segmentation that would be a natural next layer for this system if the bedside camera were replaced by a 24-hour video stream.

False-positive and false-negative behaviour are summarised below. False positives on fall arise from rapid posture changes near the floor, particularly when a person bends

over to retrieve an object. The temporal state machine partially mitigates this by requiring at least 5 percent of the window to register a fall before the composite label is emitted, but a single highly confident misdetection in the latest frame can still trigger the override path; this is the chosen behaviour, because a false positive on fall is clinically less costly than a false negative. False negatives on fall arise predominantly when the camera angle obscures the patient's head, so that the YOLOv8 detector relies on body-shape cues alone; the recommended mitigation is calibration of the camera angle at install time and a multi-camera fallback in any production deployment. False positives on drinking arise from any cup-near-face motion, such as wiping with a tissue. The recommended future mitigation is a small action-recognition model on pose sequences, similar to PoseC3D or Spatial Temporal Graph Convolutional Networks (ST-GCN), which is discussed in chapter 9. The traceability of every alert back to its frame-level facts (FR9, NFR6) means that each misclassification is attributable to a specific layer's decision, which both eases debugging and supports the clinical traceability requirement laid out by Esteva et al. (2021) and Ahmed et al. (2022).

8 Evaluation Against Research Goals

8.1 Functional Requirements

The functional requirements set out in chapter 4 are evaluated below in Table 8.1, which preserves the IDs and descriptions from Table 4.1 and adds satisfaction and explanation columns.

ID	Description	Priority	Satisfied	Explanation
FR1	Detect patient activity classes from a single frame	Must	Satisfied	Activity-classification accuracy of 91.04 percent on the 402-image validation split with the winning configuration; per-class accuracies span 83.1 percent (sit) to 100 percent (lie).
FR2	Identify bed-relative position of patient	Must	Satisfied	Implemented in bed_relation of pipeline.py with on-bed threshold of IoU at least 0.15 and side-of-bed bands at plus or minus 0.15 of bed width.
FR3	Recognise hazard and danger indicators	Must	Satisfied	Binary danger-flag accuracy of 95.52 percent on the validation split; the flag is reported as an alert independent of the activity label.
FR4	Aggregate per-frame facts into composite activity labels	Must	Satisfied	Implemented in StateMachine.infer with a priority cascade covering fall, drinking, sleeping, restless-in-bed, mobilising-with-aid, and the on-bed posture variants.
FR5	Estimate skeletal keypoints	Must	Satisfied	Pretrained YOLOv8-pose with KP_VIS at 0.4 confidence threshold gates downstream geometric reasoning.
FR6	Estimate head pose	Should	Satisfied	Five-class output (frontal, turned-left, turned-right, face-down-or-away, unknown) implemented in face_direction.
FR7	Detect cup, bottle, or wine glass	Should	Satisfied	Stock COCO YOLOv8n is run as the equipment model for these classes.
FR8	Detect mobility aids	Should	Satisfied	Walker and wheelchair are dataset classes and are flagged as boolean per-frame facts.
FR9	Emit a structured nurse-readable summary per frame	Must	Satisfied	The dictionary returned by MonitorPipeline.analyse contains all required keys plus a debug _facts payload.
FR10	Render an annotated overlay image	Should	Satisfied	draw_pose renders the skeleton and draw_summary renders the three-line text banner with alerts highlighted.

Table 8.1: Functional requirements satisfaction analysis.

8.2 Non-functional Requirements

Table 8.2 records the equivalent satisfaction analysis for the non-functional requirements.

ID	Description	Priority	Satisfied	Explanation

NFR1	End-to-end pipeline must process a frame in under 100 milliseconds on a consumer GPU	Must	Partially	The activity model alone runs comfortably above 10 frames per second on the RTX A4000 used in training, and the pose model adds roughly 30 to 40 milliseconds. Headroom on a slower consumer card such as an RTX 3060 has not been measured directly and would require a small benchmark run to confirm.
NFR2	Open-source models and libraries only	Must	Satisfied	All dependencies (Ultralytics YOLOv8, PyTorch, OpenCV, Roboflow datasets) are open-source under permissive licenses.
NFR3	No persisted personally identifiable information	Must	Satisfied	The pipeline does not write any face crops, embeddings, or identifiers to disk; only the saved weights and the model card are persisted.
NFR4	Reproducibility through fixed seed	Must	Satisfied	Random seed 42 is set in every Ultralytics training run as recorded in the args.yaml files in training_artifacts/activity*/.
NFR5	Modular pipeline	Should	Satisfied	Each layer is implemented as a separate function or class in pipeline.py with a documented contract; the activity model can be swapped by replacing model/activity_best.pt.
NFR6	Traceable safety-critical alerts	Must	Satisfied	The _facts payload returned alongside the activity label exposes every frame-level fact that contributed to the decision.
NFR7	Runnable in a Jupyter environment with a single download step	Should	Satisfied	The notebook environment-setup section runs the dataset download via the Roboflow API and resolves all hyperparameter dependencies in a single pass.

Table 8.2: Non-functional requirements satisfaction analysis.

8.3 Limitations

The system has six explicit limitations that bound the strength of the claims made in this dissertation. First, the test split of the hospital_bed dataset contains only fall and danger images, which forced all activity-classification experiments to be evaluated on the validation split rather than a held-out test split. A clean ward-collected evaluation set should be the next dataset investment for any deployment of this system. Second, the sleeping label is inferred from a closed-eye proxy that is itself a function of pose-keypoint visibility, not a true open-or-closed eye classifier. True sleep monitoring would require infrared imaging or longer observation windows. Third, the drinking label is inferred from a hand-near-mouth proxy plus a cup-or-bottle detection, which means it triggers on any cup-to-face motion such as wiping the mouth. A small action-recognition model on pose sequences such as PoseC3D or ST-GCN is the natural next step. Fourth, the side-of-bed

coordinate is in image space rather than bed space, which means that camera angle changes shift the interpretation of left and right. A clinically correct version would require two annotated bed-corner points per camera, set once at install time. Fifth, the fall-risk output is derived from the dataset's fall class rather than from a clinically validated fall-risk score such as Morse or STRATIFY; production deployment would benefit from coupling the visual signal with explicit risk scoring entered in the electronic health record. Sixth, the demonstration runs each test image independently with no tracking across frames, meaning the temporal state machine reasons per frame rather than per patient. For real video, a multi-object tracker such as ByteTrack or BoT-SORT should be added so that the state machine accumulates evidence per identity rather than per appearance.

A seventh limitation worth recording concerns generalisation. The hospital_bed dataset captures a relatively narrow visual style of ward photography, and the training set is small. The system's behaviour on a camera installed at a different angle, in a ward with different lighting, or for a patient demographic under-represented in the training set has not been measured. The recommended mitigation is targeted re-collection or hospital-specific fine-tuning before deployment, in line with the recommendations of Esteva et al. (2021) for the deployment of medical deep-learning systems. An eighth limitation is that the equipment model used for cup and bottle detection is COCO-pretrained and therefore sees the bedside scene through a generic-domain lens. A modest hospital-equipment fine-tune of this model is a low-cost extension that would tighten the drinking precondition without altering the system architecture.

A ninth limitation concerns the deterministic thresholds at the heart of the temporal state machine. The on-bed IoU threshold of 0.15, the drinking proximity threshold of 0.6, the sleeping lying-fraction threshold of 0.85, and the restless motion threshold of 0.08 were tuned by inspection on the validation split rather than learned from data. These thresholds are robust on the dataset on which they were tuned but would be expected to drift on a new camera installation. A clinically defensible deployment would replace each numeric threshold with either a calibrated learned function or a small ablation that records the sensitivity of the composite activity to plausible threshold shifts. The implementation does not currently include such a calibration pass.

A tenth limitation concerns the absence of active demographic stratification in the evaluation. The metrics in chapter 7 are reported as aggregate accuracy across the validation split. They do not split out performance by patient age, body habitus, skin tone, or assistive equipment, because the dataset annotations do not include those covariates. Esteva et al. (2021) and Litjens et al. (2017) note that systems that pass aggregate-accuracy thresholds can still exhibit unacceptable performance on specific subgroups, and the present project cannot rule that risk out. Any deployment would require demographic stratified evaluation as a prerequisite to clinical use.

9 Conclusion and Future Work

9.1 Summary

This dissertation presented an AI-based patient activity monitoring system designed as a decision-support prototype for nursing observation in hospital wards. The system is built as a five-layer perception pipeline that converts a single overhead RGB frame into a structured nurse-readable activity description with explicit safety alerts. The core detection layer is a YOLOv8 small backbone fine-tuned for 30 epochs on the hospital_bed Roboflow dataset, achieving a validation activity-classification accuracy of 91.04 percent, fall accuracy of 92 percent, and binary danger-flag accuracy of 95.52 percent. Detection mAP at the 0.50 IoU threshold reached 0.9433, and mAP at the 0.50 to 0.95 average reached 0.7801. The remaining four layers (pose estimation, bed-spatial reasoning, head-pose inference, and a temporal state machine) lift the detection output into clinically meaningful labels such as sleeping, drinking, restless-in-bed, and mobilising-with-aid. The contribution lies in the integration: a small fine-tuned hospital-domain detector paired with a generic pose head and a deterministic temporal aggregator, producing nurse-readable JSON in which every alert is traceable to a specific frame-level fact. All ten functional requirements and seven non-functional requirements set out in chapter 4 are satisfied or partially satisfied as evaluated in chapter 8. The system is delivered with reproducible weights, a model card, and a runnable notebook, and is suitable as the basis for further controlled-environment validation.

9.2 Lessons Learnt

Three lessons stand out at the close of the project. The first is that the hardest decisions in a layered perception system are not where to put the deep-learning models but where to put the deterministic glue between them. Choosing the IoU threshold for the on-bed rule, choosing the priority cascade in the temporal state machine, and choosing which keypoints to trust at what confidence are decisions that are invisible in a benchmark table but dominate the failure modes of the system in deployment. The literature has tended to focus on the deep-learning components and to leave the glue under-discussed, which is a bias worth resisting in any practical implementation.

The second lesson is that the trade-off between an end-to-end vision-language model and an explicit, layered pipeline is more interesting than it first appears. The original project brief proposed a VLM-driven reasoning layer, and the literature on large vision-language models has been moving rapidly enough (Liu et al., 2024; Wang et al., 2024) that this would have been a defensible technical choice. The choice not to adopt one was made on grounds of latency, traceability, and the small size of the activity vocabulary in scope. It would not have been the right choice for a different project, and the question of when a VLM becomes the right tool for ward monitoring deserves its own investigation.

The third lesson is that test splits matter more than headline metrics. The hospital_bed test split contains only fall and danger images, which would have invalidated any naive use of test-split accuracy as the headline metric. The decision to evaluate on the validation split was made early, but it underscored a broader point about the reliability of public benchmarks: the metrics they enable are always conditioned on the splits, and a project that ignores the splits will eventually produce a misleading number.

9.3 Future Work

Four directions for future work are most promising. The first is the addition of a vision-language reasoning layer above the deterministic state machine, used not to replace the state machine but to summarise its outputs in natural language for a nurse. Models such as Qwen2-VL (Wang et al., 2024) or BLIP-2 (Li et al., 2023) could be invoked at the granularity of an event rather than a frame, producing free-form narratives that supplement the structured JSON. This use of a VLM as a narrative layer rather than a perception layer addresses the latency and traceability concerns that ruled out a frame-by-frame VLM in the present design.

The second direction is multi-frame tracking and per-patient state. The present pipeline reasons per frame and resets the state machine between unrelated frames, which works for a static demonstration but is not how a real ward camera would be used. Adding ByteTrack or BoT-SORT identity tracking would allow the temporal state machine to accumulate evidence per patient, which is essential for sleeping, restlessness, and repositioning monitoring at clinically meaningful timescales. A graph-based extension that

captures the spatial relations between patients, beds, and equipment over time, building on the framework reviewed by Wu et al. (2021), would further extend the system's capacity to reason about ward-level dynamics.

The third direction is the optional re-identification layer that was descoped from the present project on consent grounds. With appropriate ethical review, a face-recognition or person-recognition layer using FaceNet (Schroff et al., 2018) or ArcFace (Deng et al., 2022) embeddings would allow per-patient activity histories to be assembled across multiple monitoring sessions. The same layer would enable role recognition (patient versus visitor versus staff), which is currently approximated by the largest-bounding-box heuristic and is the single largest source of ambiguity when more than one person is in frame.

A fourth direction worth flagging is the migration of the pipeline to the edge, following the smart-healthcare architecture argued for by Ahmed et al. (2022) and Martinez-Martin and del Pobil (2018). A quantised YOLOv8 nano on a Jetson-class device would close the loop between sensing and alerting without ever transmitting an image off the local hardware, which is the strongest privacy posture available to a deployed monitoring system. A complementary research direction would be a fine-tuned semantic segmentation model (Minaee et al., 2022) for the bed and the patient, replacing the bounding-box geometry in the spatial reasoning layer with pixel-accurate masks. This is a low-cost gain that would tighten the on-bed rule, particularly when patients are partially under bedclothes that occlude the lower part of the body.

A fifth and broader direction concerns clinical validation. The metrics reported in this dissertation are model-level measurements on a public dataset. They do not establish clinical utility, and they cannot. Clinical utility is a separate question that asks whether the system, when deployed in a real ward, reduces missed events, reduces nurse cognitive load, and improves patient outcomes. Answering that question requires a controlled clinical study with ethical approval, a clear primary endpoint, and a pre-registered analysis plan. The present project produces an artefact that could be the perception backbone of such a study, but it does not itself constitute one. Any future work proposal that frames

the next phase of the project as a clinical study rather than a model-improvement exercise would, in my view, sit in the place where the largest unmet research need is concentrated.

References

Ahmed, I., Jeon, G., & Piccialli, F. (2022). A deep-learning-based smart healthcare system for patient's discomfort detection at the edge of Internet of Things. IEEE Internet of Things Journal, 8(13), 10318-10326.

Akbari, A., Wu, J., Grimsley, R., & Jafari, R. (2020). Hierarchical signal segmentation and classification for accurate activity recognition. IEEE Journal of Biomedical and Health Informatics, 24(7), 1990-2000.

Alzubaidi, L., Zhang, J., Humaidi, A. J., Al-Dujaili, A., Duan, Y., Al-Shamma, O., Santamaria, J., Fadhel, M. A., Al-Amidie, M., & Farhan, L. (2021). Review of deep learning: Concepts, CNN architectures, challenges, applications, future directions. Journal of Big Data, 8(1), 53.

Bochkovskiy, A., Wang, C.-Y., & Liao, H.-Y. M. (2020). YOLOv4: Optimal speed and accuracy of object detection. arXiv preprint. (Peer-reviewed version: Wang, C.-Y., Bochkovskiy, A., & Liao, H.-Y. M. (2023). YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 7464-7475.)

Chen, K., Gabriel, P., Alasfour, A., Gong, C., Doyle, W. K., Devinsky, O., Friedman, D., Dugan, P., Melloni, L., Thesen, T., Gonda, D., Sattar, S., Wang, S., & Gilja, V. (2018). Patient-specific pose estimation in clinical environments. IEEE Journal of Translational Engineering in Health and Medicine, 6, 2101111.

Deng, J., Guo, J., Yang, J., Xue, N., Kotsia, I., & Zafeiriou, S. (2022). ArcFace: Additive angular margin loss for deep face recognition. IEEE Transactions on Pattern Analysis and Machine Intelligence, 44(10), 5962-5979.

Diraco, G., Rescio, G., Siciliano, P., & Leone, A. (2023). Review on human action recognition in smart living: Sensing technology, multimodality, real-time processing, interoperability, and resource-constrained processing. Sensors, 23(11), 5281.

Esteva, A., Chou, K., Yeung, S., Naik, N., Madani, A., Mottaghi, A., Liu, Y., Topol, E., Dean, J., & Socher, R. (2021). Deep learning-enabled medical computer vision. npj Digital Medicine, 4(1), 5.

Gupta, N., Gupta, S. K., Pathak, R. K., Jain, V., Rashidi, P., & Suri, J. S. (2022). Human activity recognition in artificial intelligence framework: A narrative review. Artificial Intelligence Review, 55(6), 4755-4808.

Haque, A., Milstein, A., & Fei-Fei, L. (2020). Illuminating the dark spaces of healthcare with ambient intelligence. Nature, 585(7824), 193-202.

Jiang, P., Ergu, D., Liu, F., Cai, Y., & Ma, B. (2022). A review of YOLO algorithm developments. Procedia Computer Science, 199, 1066-1073.

Kong, Y., & Fu, Y. (2022). Human action recognition and prediction: A survey. International Journal of Computer Vision, 130(5), 1366-1401.

Li, J., Li, D., Savarese, S., & Hoi, S. (2023). BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. Proceedings of Machine Learning Research, 202, 19730-19742.

Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., van der Laak, J. A. W. M., van Ginneken, B., & Sanchez, C. I. (2017). A survey on deep learning in medical image analysis. Medical Image Analysis, 42, 60-88.

Liu, H., Li, C., Wu, Q., & Lee, Y. J. (2024). Visual instruction tuning. Advances in Neural Information Processing Systems, 36, 34892-34916.

Martinez-Martin, E., & del Pobil, A. P. (2018). Personal robot assistants for elderly care: An overview. Personal Assistants: Emerging Computational Technologies, 132, 77-91.

Minaee, S., Boykov, Y., Porikli, F., Plaza, A., Kehtarnavaz, N., & Terzopoulos, D. (2022). Image segmentation using deep learning: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence, 44(7), 3523-3542.

Mukherjee, D., Mondal, R., Singh, P. K., Sarkar, R., & Bhattacharjee, D. (2020). EnsemConvNet: A deep learning approach for human activity recognition using

smartphone sensors for healthcare applications. Multimedia Tools and Applications, 79(41), 31663-31690.

Pareek, P., & Thakkar, A. (2021). A survey on video-based human action recognition: Recent updates, datasets, challenges, and applications. Artificial Intelligence Review, 54(3), 2259-2322.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning transferable visual models from natural language supervision. Proceedings of Machine Learning Research, 139, 8748-8763.

Redmon, J., & Farhadi, A. (2018). YOLOv3: An incremental improvement. arXiv preprint. IEEE Transactions on Pattern Analysis and Machine Intelligence

Schroff, F., Kalenichenko, D., & Philbin, J. (2018). FaceNet: A unified embedding for face recognition and clustering. IEEE Transactions on Pattern Analysis and Machine Intelligence, 40(6), 1480-1493.

Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. Journal of Big Data, 6(1), 60.

Singh, R., Kushwaha, A. K. S., Srivastava, R., & Kushwaha, A. K. S. (2019). Multi-view recognition system for human activity based on multiple features for video surveillance system. Multimedia Tools and Applications, 78(12), 17165-17196.

Terven, J., Cordova-Esparza, D.-M., & Romero-Gonzalez, J.-A. (2023). A comprehensive review of YOLO architectures in computer vision: From YOLOv1 to YOLOv8 and YOLO-NAS. Machine Learning and Knowledge Extraction, 5(4), 1680-1716.

Vrigkas, M., Nikou, C., & Kakadiaris, I. A. (2015). A review of human activity recognition methods. Frontiers in Robotics and AI, 2, 28.

Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., Fan, Y., Dang, K., Du, M., Ren, X., Men, R., Liu, D., Zhou, C., Zhou, J., & Lin, J. (2024). Qwen2-VL: Enhancing vision-language model’s perception of the world at any resolution.

arXiv preprint. (Companion peer-reviewed work: Bai, J., et al. (2023). Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond. arXiv preprint arXiv:2308.12966.)

Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., & Yu, P. S. (2021). A comprehensive survey on graph neural networks. IEEE Transactions on Neural Networks and Learning Systems, 32(1), 4-24.

Zhao, Z.-Q., Zheng, P., Xu, S.-T., & Wu, X. (2019). Object detection with deep learning: A review. IEEE Transactions on Neural Networks and Learning Systems, 30(11), 3212-3232.

Zou, Z., Chen, K., Shi, Z., Guo, Y., & Ye, J. (2023). Object detection in 20 years: A survey. Proceedings of the IEEE, 111(3), 257-276.

Appendices

Appendix 1: Codebase

https://portdotacdotuk-my.sharepoint.com/:f:/g/personal/up2310520_myport_ac_uk/IgB5EnJHGBx7TZ3i2skYd3JsASnWYp1HYUuS5TpywC1dq18?e=ntgqhx

Appendix 2: Project Specification

Project Specification

1. Basic details

Student name:	Manoj Navarathinarasa
Draft project title:	Agentic Vision-Language System for Domain-Specific Object Detection and Activity Monitoring in Hospital Ward Environments
Course and year:	2025 - 2026
Client organisation:	N/A (Academic Project)
Client contact name:	Rinat Khusainov
Project supervisor:	Rinat Khusainov

Outline of the project environment

Hospital wards are complex indoor environments containing a wide variety of medical equipment, furniture, and individuals including patients, staff, and visitors. Effective monitoring of these environments is critical for ensuring patient safety, maintaining hygiene standards, and enabling timely clinical responses. Current monitoring in most hospital wards relies heavily on manual observation by nursing staff, which is resource-intensive, inconsistent, and prone to human error, particularly during overnight shifts or periods of understaffing.

Recent advances in deep learning-based object detection have demonstrated strong capability in identifying and locating objects within images. The Hospital Indoor Object Detection (HIOD) dataset by Hu, Li and Wang (2023) established a benchmark for detecting 56 object categories across 4,417 annotated hospital images. However, detection alone is insufficient knowing that a "person" and a "bed" exist in a frame does not answer whether the patient is resting, active, or in distress. There is a clear gap between low-level object detection and high-level scene understanding that requires contextual reasoning.

This project addresses that gap by building an intelligent monitoring system that combines domain-specific object detection with agentic vision-language reasoning, enabling the system to not only detect objects but also interpret activities, assess room conditions, and monitor patient behaviour in hospital ward environments.

The problem to be solved

The core problem is that existing object detection systems for hospital environments operate at the detection level only they identify what objects are present but do not reason about the relationships between objects, the state of the environment, or the activities being performed. For example, detecting a "person" and a "water bottle" as separate objects provides no insight into whether the patient is drinking water, which may be clinically relevant for monitoring fluid intake.

Aims: The aim of this project is to design and implement a domain-specific intelligent monitoring system for hospital ward environments that combines object detection with vision-language reasoning to provide actionable scene understanding.

Objectives:

1. Fine-tune a state-of-the-art object detection model (YOLOv8/YOLOv6) on hospital-specific objects using the HIOD dataset and supplementary custom annotations.
2. Implement an agentic vision-language reasoning layer using an open-source Vision-Language Model (such as LLaVA or Qwen2-VL) that interprets spatial relationships between detected objects to perform activity recognition.
3. Develop activity monitoring capabilities including: detecting a person drinking water, assessing whether a patient is on the bed or active, and evaluating overall room condition.
4. Integrate person identification using face/person re-identification techniques to associate activities with specific individuals.
5. Build a real-time monitoring dashboard that displays detection results, activity classifications, room condition scores, and alerts.
6. Evaluate the system both quantitatively (detection mAP, activity classification accuracy) and qualitatively (usability, clinical relevance).

Constraints:

- The project must use publicly available or requestable datasets (HIOD) and open-source models to ensure reproducibility.
- Real patient data will not be used; all testing will be performed on publicly available hospital ward images/videos or simulated environments.
- Computational resources are limited to consumer-grade GPUs, so model selection must balance accuracy with inference efficiency.
- The system must operate at near real-time speeds (minimum 10 FPS) to be practical for monitoring applications.

Breakdown of tasks

Phase 1: Research and Foundation (Weeks 1-3)

- Conduct a comprehensive literature review on hospital object detection, vision-language models, activity recognition, and agentic AI systems.
- Study the HIOD dataset structure, annotation format, and benchmark results in depth.
- Review state-of-the-art VLMs (LLaVA, Qwen2-VL, InternVL) and select the most suitable model based on accuracy-efficiency tradeoff.
- Identify gaps in existing approaches that justify the two-layer architecture.

Phase 2: Dataset Preparation and Object Detection (Weeks 3-6)

- Request and obtain the HIOD dataset from the authors.
- Prepare supplementary annotations for activity-related objects (water bottles, cups, food trays) if not covered by HIOD.
- Convert annotations to the required format (YOLO txt format from Pascal VOC XML).
- Fine-tune YOLOv8 on the HIOD dataset and evaluate detection performance (mAP, AP50, AP75).
- Benchmark against the results reported by Hu et al. (2023) to validate the detection layer.

Phase 3: Vision-Language Agent Development (Weeks 5-9)

- Design the structured detection context builder that converts YOLO outputs into natural language scene descriptions.
- Implement the agentic reasoning pipeline using a VLM to classify activities from detection context plus raw frame input.
- Develop prompt engineering strategies for each monitoring task: drinking detection, room condition assessment, patient bed/activity status, and general activity classification.
- Implement person re-identification using ArcFace or FaceNet embeddings for associating activities with individuals.

Phase 4: System Integration and Dashboard (Weeks 8-11)

- Integrate the object detection layer with the VLM reasoning layer into a unified pipeline.
- Build a real-time monitoring dashboard using Streamlit or Gradio.
- Implement alert mechanisms for key events (patient leaving bed, room condition deterioration, drinking activity detected).
- Optimise the end-to-end pipeline for near real-time performance.

Phase 5: Evaluation and Documentation (Weeks 10-15)

- Conduct quantitative evaluation: detection mAP, activity classification accuracy, precision/recall for each monitoring task, inference speed.
- Perform qualitative evaluation: assess system interpretability, reasoning quality, and practical usability.

- Compare the two-layer approach against detection-only baselines to demonstrate the value of the reasoning layer.
- Write the dissertation, prepare the demonstration, and finalise all deliverables.

Skills required and acquisition plan:

Skill	Acquisition Plan
Object detection (YOLO)	HIOD repo codebase, Ultralytics docs
Vision-Language Models	HuggingFace tutorials, LLaVA/Qwen documentation
Prompt engineering for VLMs	Experimentation + literature review
Face/person re-identification	ArcFace/FaceNet tutorials, InsightFace library
Streamlit/Gradio dashboards	Official documentation, prior experience
Academic writing	Supervisor feedback, iterative drafting

Project deliverables

Deliverable	Description
Literature Review	Comprehensive review of hospital object detection, VLMs, activity recognition, and agentic AI systems
Fine-tuned Object Detection Model	YOLOv8/YOLOv6 model fine-tuned on HIOD dataset with benchmark comparison
Agentic VLM Reasoning Pipeline	Implemented vision-language agent that interprets detected objects for activity monitoring
Integrated Monitoring System	End-to-end system combining detection, reasoning, person identification, and alerting
Real-time Dashboard	Streamlit/Gradio web application for live monitoring and visualisation
Evaluation Report	Quantitative and qualitative evaluation comparing the proposed approach against baselines
Dissertation Document	Complete MSc dissertation following university guidelines
Source Code Repository	Well-documented codebase with installation and usage instructions
Project Demonstration	Live demonstration of the system to the supervisor and examiners

Requirements

Functional Requirements:

ID	Requirement	Priority
FR1	The system shall detect and classify hospital ward objects from camera input with mAP ≥ 0.45	Must Have
FR2	The system shall detect when a person is drinking water by reasoning about person-object spatial relationships	Must Have
FR3	The system shall assess room condition (good/moderate/poor) based on detected object arrangement and state	Must Have
FR4	The system shall determine whether a patient is on the bed or active (standing/walking)	Must Have
FR5	The system shall identify individuals using face/person re-identification	Should Have
FR6	The system shall provide real-time activity monitoring with alerts for key events	Should Have
FR7	The system shall display results on an interactive dashboard with live video feed	Should Have
FR8	The system shall log all detected activities with timestamps for retrospective review	Could Have

Non-Functional Requirements:

ID	Requirement	Priority
NFR1	The system shall process frames at a minimum of 10 FPS for near real-time operation	Must Have
NFR2	The system shall run on a consumer-grade GPU (e.g., NVIDIA RTX 3060 or equivalent)	Must Have
NFR3	The system shall use only open-source models and libraries	Must Have
NFR4	The system shall not store or transmit any personally identifiable information beyond the local session	Must Have
NFR5	The system interface shall be accessible via a standard web browser	Should Have

Requirements will be further refined through iterative discussions with the project supervisor during supervision meetings.

Legal, ethical, professional, social issues

Ethical Considerations: This project involves processing images that may contain human faces and activities. However, no real patient data will be collected or used.

All development and testing will use the publicly available HIOD dataset (which has been through its own ethical approval at the University of Tennessee) and supplementary images from open-source datasets or simulated environments. No hospital visits, patient interactions, or clinical data collection are involved.

Data Protection and Privacy: The system will be designed with privacy by default. Person re-identification will use locally computed embeddings only and will not connect to any external database. No images or personal data will be stored persistently or transmitted externally. The system complies with GDPR principles of data minimisation and purpose limitation.

Research Ethics Approval: Since the project uses only publicly available datasets and does not involve human participants, clinical environments, or collection of personal data, formal ethics approval is not anticipated to be required. However, this will be confirmed with the supervisor and the university ethics committee at the outset of the project. If any component requires ethics review, the application will be submitted promptly.

Social Impact: The system has the potential to support nursing staff by providing an additional monitoring layer, reducing workload during high-demand periods. However, it is designed as an assistive tool, not a replacement for human judgement. Concerns around surveillance and patient autonomy are acknowledged, and any future deployment would require patient consent frameworks and institutional review.

Facilities and resources

Resource	Details
Hardware	Personal laptop/workstation with NVIDIA GPU (RTX 3060 or above) for local model training and inference
Cloud GPU (if needed)	Google Colab Pro / University HPC cluster for intensive training runs
HIOD Dataset	4,417 annotated hospital images, 56 categories
Open-source Models	YOLOv8 (Ultralytics), LLaVA/Qwen2-VL (HuggingFace), ArcFace (InsightFace)
Development Tools	Python 3.10+, PyTorch, OpenCV, LangChain, Streamlit/Gradio, Jupyter
University Library	Access to IEEE, Elsevier, Springer for literature review
Test Videos	Open-source hospital ward footage from YouTube/public datasets for system testing

No additional funding is required. All tools and models are open-source. The only external dependency is the HIOD dataset, for which a request will be submitted immediately using the university email

Project plan

GANTT CHART

Task	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9	Wk 10	Wk 11	Wk 12	Wk 13	Wk 14	Wk 15	Wk 16
Literature Review	▌	▌	▌	▌	▌	▌	▌	▌	▌							
HIOD Dataset Request & Setup	▌	▌														
Dataset Preparation & Conversion			▌	▌												
YOLO v8 Fine-tuning on HIOD				▌	▌	▌										
Detection Layer Evaluation						▌	▌									
VLM Selection & Setup					▌	▌										
Agentic Reasoning Pipeline Dev							▌	▌	▌							
Person Re-ID Integration									▌	▌						

System Integration								▮	▮	▮	▮					
Dashboard Development										▮	▮	▮				
Testing & Evaluation											▮	▮	▮			
Dissertation Writing							▮	▮	▮	▮	▮	▮	▮	▮	▮	▮
Final Review & Submission Prep																

Risk Assessment:

Risk	Likelihood	Impact	Mitigation
HIOD dataset access delayed	Medium	High	Begin development with COCO hospital-relevant subset; follow up with authors weekly
VLM inference too slow for real-time	Medium	Medium	Use quantised models (4-bit); process every Nth frame; use smaller VLM variants
GPU resources insufficient for training	Low	Medium	Use Google Colab Pro or university HPC; reduce batch size; use pre-trained weights with minimal fine-tuning
Activity classification accuracy too low	Medium	Medium	Supplement with rule-based heuristics for spatial relationship analysis; increase prompt engineering iterations
Scope creep from professor's extended requirements	Medium	Low	Prioritise MoSCoW requirements; defer "Could Have" features if timeline is tight

Supervisor unavailability during key periods	Low	Medium	Maintain async communication via email; prepare specific questions in advance for each meeting

Supervision meetings

Weekly or fortnightly supervision meetings have been agreed with Supervisor throughout January-May 2026, primarily via Microsoft Teams, with face-to-face meetings at the University of Portsmouth when possible. Email will be used for interim updates. During supervisor absences, progress will continue per the project plan with email communication maintained.

Project mode

If there are two possibilities for your project mode, after negotiation, please record your planned duration and submission date. It is also helpful to record your initial registration mode (i.e. are you a full time or a part time student). Remember, the exact dates will be announced through Moodle these represent a generic guideline.

	Please delete as appropriate	
Registration mode	Full Time	Part Time
Project mode	Full Time	Part Time
Planned submission deadline	10/5/26	

Signatures

	Signature:	Date:
Student	Manoj Navarathinarasa	01-02-2026
Client	Rinat Khusainov	01-02-2026
Project supervisor	Rinat Khusainov	01-02-2026

Appendix 3: Ethic Application

UNIVERSITY OF PORTSMOUTH

Certificate of Ethics Review

Project title: Agentic Vision-Language System for Domain-Specific Object Detection and Activity Monitoring in Hospital Ward Environments

Name:	Manoj Navarathinarasa	**User ID**:	UP2310520	**Application date**:	23/02/2026 21:52:44	**ER Number**:	TETHIC-2026-113067

You must download your referral certificate, print a copy and keep it as a record of this review.

The FEC representative(s) for the **School of Computing** is/are **Elisavet Andrikopoulou, Kirsten Smith**

It is your responsibility to follow the University Code of Practice on Ethical Standards and any Department/School or professional guidelines in the conduct of your study including relevant guidelines regarding health and safety of researchers including the following:

- University Policy
- Safety on Geological Fieldwork

It is also your responsibility to follow University guidance on Data Protection Policy:

- General guidance for all data protection issues
- University Data Protection Policy

Which school/department do you belong to?: **School of Computing**
What is your primary role at the University?: **Postgraduate Student**
What is the name of the member of staff who is responsible for supervising your project?: **Rinat Khusainov**
Is the study likely to involve human subjects (observation) or participants?: No
Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?: No
Are there risks of significant damage to physical and/or ecological environmental features?: No
Are there risks of significant damage to features of historical or cultural heritage (e.g. impacts of study techniques, taking of samples)?: No
Does the project involve animals in any way?: No
Could the research outputs potentially be harmful to third parties?: No
Could your research/artefact be adapted and be misused?: No
Will your project or project deliverables be relevant to defence, the military, police or other security organisations and/or in addition, could it be used by others to threaten UK security?: No
Please read and confirm that you agree with the following statements: I confirm that I have considered the implications for data collection and use, taking into consideration legal requirements (UK GDPR, Data Protection Act 2018 etc.), I confirm that I have considered the impact of this work and and taken any reasonable action to mitigate potential misuse of the project outputs, I confirm that I will act ethically and honestly throughout this project

Supervisor Review

As supervisor, I will ensure that this work will be conducted in an ethical manner in line with the University Ethics Policy.

Supervisor comments:

Supervisor's Digital Signature: **rinat.khusainov@port.ac.uk** Date: **23/02/2026**